#1 · procedures_periprocedural

Glasgow-Blatchford score

Answer

A pre-endoscopy risk score using only clinical and lab data. A score of zero identifies patients at very low risk who can often be managed as outpatients without urgent endoscopy.

#2 · procedures_periprocedural

Restrictive transfusion threshold

Answer

For most stable patients without cardiac disease, transfuse at a hemoglobin around 7 g/dL. Patients with cardiovascular disease may use a slightly higher threshold near 8 g/dL.

#3 · procedures_periprocedural

Why liberal transfusion harms variceal bleeders

Answer

Overtransfusion raises portal venous pressure, which increases the risk of rebleeding in patients with variceal hemorrhage.

#4 · procedures_periprocedural

Timing of endoscopy in upper GI bleeding

Answer

Endoscopy within 24 hours after resuscitation is standard for most patients. Very early endoscopy within 6 hours has not shown benefit in unselected stable patients.

#5 · procedures_periprocedural

Octreotide in variceal bleeding

Answer

Reduces splanchnic blood flow and portal pressure. Started immediately in suspected variceal hemorrhage and continued for 3 to 5 days.

#6 · procedures_periprocedural

Antibiotics in cirrhotic GI bleeders

Answer

Prophylactic antibiotics such as ceftriaxone reduce mortality and rebleeding in cirrhotic patients with GI bleeding, regardless of whether ascites is present.

#7 · procedures_periprocedural

Obscure GI bleeding workup order

Answer

After negative upper endoscopy and colonoscopy, video capsule endoscopy is the next step. Deep enteroscopy follows if a lesion is found needing biopsy or therapy.

#8 · procedures_periprocedural

Meckel scan

Answer

Technetium-99m pertechnetate scan used to detect ectopic gastric mucosa, most useful in young patients with painless lower GI bleeding after a negative standard workup.

#9 · procedures_periprocedural

Perforation risk: EGD vs colonoscopy

Answer

Diagnostic EGD carries under 1 in 10,000 perforation risk. Diagnostic colonoscopy carries roughly 1 in 1,000 to 3,000, higher with polypectomy.

#10 · procedures_periprocedural

Most common serious colonoscopy complication

Answer

Post-polypectomy bleeding, either immediate (managed with clips or thermal therapy) or delayed up to two weeks later.

#11 · procedures_periprocedural

Post-polypectomy syndrome

Answer

Localized pain, fever, and leukocytosis without free air after electrocautery polypectomy, from a transmural burn. Managed conservatively with bowel rest, fluids, and antibiotics.

#12 · procedures_periprocedural

Post-ERCP pancreatitis risk factors

Answer

Suspected sphincter of Oddi dysfunction and difficult cannulation are the strongest risk factors. Occurs in roughly 3 to 10 percent of ERCPs overall.

#13 · procedures_periprocedural

Preventing post-ERCP pancreatitis

Answer

Rectal indomethacin given immediately before or after the procedure in moderate to high risk patients, plus prophylactic pancreatic duct stenting in high risk difficult cannulation cases.

#14 · procedures_periprocedural

Leading cause of endoscopy related mortality

Answer

Cardiopulmonary events related to sedation, not bleeding or perforation.

#15 · procedures_periprocedural

Radiation safety: distance principle

Answer

Radiation intensity follows the inverse square law with distance, so doubling distance from the source cuts exposure to about one quarter.

#16 · procedures_periprocedural

Fluoroscopy PPE

Answer

Lead apron, thyroid shield, and leaded eyewear protect against scatter radiation during procedures like ERCP.

#17 · procedures_periprocedural

Left lateral decubitus rationale

Answer

Allows secretions and refluxed material to pool away from the vocal cords, protecting the airway during sedated upper endoscopy.

#18 · procedures_periprocedural

Pregnancy positioning

Answer

Left lateral tilt avoids inferior vena cava compression by the gravid uterus, which can otherwise cause maternal hypotension when supine.

#19 · procedures_periprocedural

Sedation in special populations

Answer

Pregnant patients: lowest effective dose, favorable fetal safety profile. Elderly patients: more conservative dosing given reduced clearance and cardiac reserve.

#20 · procedures_periprocedural

Toxic megacolon and endoscopy

Answer

Full colonoscopy with standard insufflation is avoided due to perforation risk in a friable, distended colon. A limited unprepped flexible sigmoidoscopy with minimal air is preferred if assessment is needed.

#21 · procedures_periprocedural

Urgent biliary drainage timing

Answer

Acute cholangitis with sepsis requires biliary decompression, generally within 24 hours, since delay significantly increases mortality.

#22 · procedures_periprocedural

Therapeutic gastroscope selection

Answer

Larger working channel allows better suction of blood and clot and easier passage of hemostatic devices, preferred when active bleeding is suspected.

#23 · procedures_periprocedural

Cap assisted / wide angle colonoscopy

Answer

Improves mucosal visualization behind folds and flexures, useful in patients with a history of missed lesions.

#24 · procedures_periprocedural

Echoendoscope FNA use

Answer

Provides real time ultrasound guided sampling of subepithelial lesions that lie beneath the mucosa, improving yield over standard forceps biopsy.

#25 · procedures_periprocedural

Cold snare vs hot snare

Answer

Cold snare is now preferred for polyps 6 to 9mm, achieving similar complete resection rates with lower bleeding risk than hot snare.

#26 · procedures_periprocedural

Jumbo biopsy forceps indication

Answer

Used when standard forceps repeatedly yield only superficial tissue, such as suspected eosinophilic esophagitis needing deeper sampling.

#27 · procedures_periprocedural

Sessile serrated lesion appearance

Answer

Flat, subtle contour, often covered by a mucus cap, making it easy to miss without careful inspection. More common in the proximal colon.

#28 · procedures_periprocedural

Paris classification depressed component

Answer

A depressed morphology carries a higher risk of submucosal invasion even in small lesions, influencing resection approach.

#29 · procedures_periprocedural

Adenoma detection rate (ADR)

Answer

The most validated colonoscopy quality indicator. Minimum benchmark is at least 25 percent overall, with sex specific targets. Higher ADR correlates with lower interval cancer rates.

#30 · procedures_periprocedural

Cecal intubation rate benchmark

Answer

Minimum accepted benchmark is at least 95 percent for screening colonoscopy.

#31 · procedures_periprocedural

Withdrawal time benchmark

Answer

At least 6 minutes during a normal screening colonoscopy, associated with higher adenoma detection.

#32 · procedures_periprocedural

Surveillance: 1-2 small tubular adenomas

Answer

7 to 10 year interval, similar to average risk screening.

#33 · procedures_periprocedural

Surveillance: adenoma with villous features or HGD

Answer

3 year interval, even without other high risk features.

#34 · procedures_periprocedural

Surveillance: 5+ adenomas or piecemeal resection

Answer

Shortened interval, often 3 years or less, with an earlier repeat exam at a piecemeal resection site to confirm complete removal.

#35 · procedures_periprocedural

Serrated polyposis syndrome

Answer

Defined by cumulative polyp count and size criteria. Moves the patient to a much shorter surveillance interval given elevated cumulative cancer risk.

#36 · procedures_periprocedural

Low vs high bleeding risk procedures

Answer

Low risk: diagnostic endoscopy with or without biopsy, no anticoagulation interruption needed. High risk: large polypectomy, ERCP with sphincterotomy, PEG placement, generally requiring interruption.

#37 · procedures_periprocedural

Warfarin management before high risk procedures

Answer

Hold approximately 5 days before, check INR the day of the procedure to confirm normalization.

#38 · procedures_periprocedural

Bridging anticoagulation

Answer

Reserved for genuinely high thrombotic risk patients, such as a mechanical mitral valve or recent VTE. Routine bridging in lower risk atrial fibrillation increases bleeding without proportional benefit.

#39 · procedures_periprocedural

DOAC periprocedural hold

Answer

Generally held 1 to 2 days for low risk procedures, with duration influenced by renal function since it affects drug clearance.

#40 · procedures_periprocedural

Aspirin for secondary prevention around endoscopy

Answer

Generally continued through procedures including polypectomy, since the cardiovascular risk of stopping usually outweighs the modest bleeding risk reduction.

#41 · procedures_periprocedural

Dual antiplatelet therapy after coronary stenting

Answer

Elective endoscopy is generally delayed, or managed in close coordination with cardiology, until the minimum recommended dual therapy duration has passed.

#42 · procedures_periprocedural

Antibiotic prophylaxis indications

Answer

PEG tube placement (reduces peristomal infection) and ERCP with expected incomplete biliary drainage (such as a hilar stricture in PSC).

#43 · procedures_periprocedural

Antibiotic prophylaxis NOT needed

Answer

Routine colonoscopy or diagnostic EGD, even in patients with prosthetic joints or valves, since endoscopy related bacteremia rarely causes clinically significant infection.

#44 · procedures_periprocedural

GLP-1 receptor agonists and sedation

Answer

Delay gastric emptying, raising aspiration risk. Consider holding the medication or extending the fasting interval before elective sedated procedures.

#45 · procedures_periprocedural

Airway assessment before sedation

Answer

Uses a structured classification system evaluating mouth opening, tongue size, and neck mobility to predict a difficult airway.

#46 · procedures_periprocedural

Most common sedation adverse event

Answer

Hypoxemia from over sedation, more common than hypotension or cardiac arrhythmia.

#47 · procedures_periprocedural

Sedation reversal agents

Answer

Flumazenil reverses benzodiazepines. Naloxone reverses opioids. Propofol has no specific reversal agent and requires supportive airway management.

#48 · procedures_periprocedural

Capnography benefit

Answer

Detects hypoventilation earlier than pulse oximetry detects oxygen desaturation, giving an earlier warning during sedation.

#49 · procedures_periprocedural

Stepwise airway management for oversedation

Answer

Stop the procedure, apply supplemental oxygen, reposition the airway, then escalate to bag mask ventilation and advanced airway management if needed.

#50 · procedures_periprocedural

PEG tube indication

Answer

Long term enteral nutrition anticipated to last more than 4 to 6 weeks, with a functional GI tract.

#51 · procedures_periprocedural

Confirming safe PEG access

Answer

Transillumination with a clear finger indentation (one two three) sign confirms the stomach is apposed to the abdominal wall without an interposed organ.

#52 · procedures_periprocedural

PEG-J or direct percutaneous jejunostomy

Answer

Used when gastric feeding is contraindicated, such as severe gastroparesis or recurrent aspiration despite gastric feeding.

#53 · procedures_periprocedural

Buried bumper syndrome

Answer

Internal bumper erodes into the gastric or abdominal wall. Presents with difficulty flushing, pain, and resistance to tube rotation. Needs endoscopic or surgical correction.

#54 · procedures_periprocedural

Early PEG tube dislodgement

Answer

Within the first 7 to 10 days before the tract matures. A true emergency, since blind replacement risks a false tract and peritoneal leak. Needs endoscopic or fluoroscopic guidance.

#55 · procedures_periprocedural

Informed consent components

Answer

Indication, procedure details, reasonable alternatives, and material risks specific to the patient, including sedation risks. A signed form alone is not sufficient.

#56 · procedures_periprocedural

Goals of care before PEG in advanced dementia

Answer

Should reflect the patient's previously expressed wishes or surrogate judgment. Evidence does not show a mortality or aspiration benefit over careful hand feeding.

#57 · procedures_periprocedural

Decision-specific capacity

Answer

Capacity is assessed per decision, not globally. A patient may retain capacity for a simple decision while lacking it for a complex one.

#58 · procedures_periprocedural

Implied consent in emergencies

Answer

When a patient lacks capacity, has no available surrogate, and delay would risk life or limb, emergency procedures can proceed under implied consent.

#59 · procedures_periprocedural

Button battery ingestion

Answer

Esophageal button batteries need emergent removal, generally within about 2 hours, since they generate an electrical current that can burn through tissue quickly.

#60 · procedures_periprocedural

Sharp esophageal foreign bodies

Answer

Require urgent removal given elevated perforation risk compared with blunt objects.

#61 · procedures_periprocedural

Esophageal perforation diagnosis

Answer

Chest pain, subcutaneous emphysema, and fever after instrumentation. Confirmed with contrast esophagram or CT showing extraluminal contrast or air.

#62 · procedures_periprocedural

Esophageal perforation management

Answer

Small, early, contained perforations may be managed endoscopically with clips or stents. Larger, later presentations with mediastinal contamination often need surgery.

#63 · procedures_periprocedural

Charcot's triad and Reynolds' pentad

Answer

Charcot's triad: fever, jaundice, RUQ pain. Reynolds' pentad adds hypotension and altered mental status, signaling severe cholangitis.

#64 · procedures_periprocedural

Sigmoid volvulus first line treatment

Answer

Endoscopic detorsion with a flexible sigmoidoscope, leaving a rectal tube afterward, in the absence of peritonitis or ischemia.

#65 · procedures_periprocedural

Contraindications to endoscopic detorsion

Answer

Peritoneal signs or suspected ischemia mandate emergency surgical evaluation instead of endoscopic reduction.

#66 · esophageal_disorders

What is the lower esophageal sphincter made of, and why does that matter clinically?

Answer

It is smooth muscle, unlike the upper esophageal sphincter which is striated muscle. This is why smooth muscle diseases like scleroderma and achalasia mainly affect the lower esophagus, while neuromuscular diseases like myasthenia gravis mainly affect the upper esophagus and pharynx.

#67 · esophageal_disorders

Define Barrett esophagus.

Answer

Barrett esophagus is replacement of the normal squamous lining of the distal esophagus by columnar epithelium with intestinal metaplasia, confirmed on biopsy. It develops from chronic acid reflux and carries a small yearly risk of progressing to dysplasia and adenocarcinoma.

#68 · esophageal_disorders

What is the surveillance interval for non-dysplastic Barrett esophagus?

Answer

Every three to five years. Shorter intervals are used once dysplasia is confirmed, and endoscopic ablation is considered for low grade dysplasia and beyond.

#69 · esophageal_disorders

How is high grade dysplasia or mucosal cancer in Barrett esophagus typically treated now?

Answer

With endoscopic mucosal resection of visible lesions combined with ablation of the remaining Barrett segment, which has largely replaced esophagectomy for disease confined to the mucosa.

#70 · esophageal_disorders

What are the main risk factors for Barrett esophagus and esophageal adenocarcinoma?

Answer

Long duration GERD, male sex, age over fifty, central obesity, smoking, and a family history of Barrett esophagus or esophageal cancer.

#71 · esophageal_disorders

What is the classic endoscopic appearance of eosinophilic esophagitis?

Answer

Concentric rings sometimes called trachealization, longitudinal furrows, and whitish exudates representing eosinophil aggregates, sometimes with a narrow caliber esophagus in severe cases.

#72 · esophageal_disorders

What histologic threshold is used to diagnose eosinophilic esophagitis?

Answer

At least fifteen eosinophils per high power field on biopsy from both the proximal and distal esophagus, after other causes of esophageal eosinophilia have been excluded.

#73 · esophageal_disorders

Who typically develops eosinophilic esophagitis, and what is the classic presenting symptom?

Answer

Younger patients, often men in their twenties or thirties with a personal history of atopy such as asthma or eczema. The classic symptom is solid food dysphagia with recurrent food impaction.

#74 · esophageal_disorders

Name the main treatment options for eosinophilic esophagitis.

Answer

Topical swallowed corticosteroids, proton pump inhibitor therapy, elimination diets removing common triggers like dairy and wheat, biologic therapy for refractory cases, and careful dilation for strictures.

#75 · esophageal_disorders

What is an inlet patch?

Answer

A congenital island of heterotopic gastric mucosa found just below the upper esophageal sphincter, usually discovered incidentally and almost always asymptomatic.

#76 · esophageal_disorders

Give two causes of extrinsic esophageal compression.

Answer

An enlarged left atrium and an aberrant right subclavian artery are two classic causes. Others include aortic aneurysm and mediastinal masses. Dilation does not fix compression from an external structure.

#77 · esophageal_disorders

What is a Schatzki ring and what is it associated with?

Answer

A thin mucosal ring at the squamocolumnar junction, strongly associated with hiatal hernia, causing intermittent solid food dysphagia often called steakhouse syndrome.

#78 · esophageal_disorders

What is Plummer-Vinson syndrome and what long term risk does it carry?

Answer

The triad of iron deficiency anemia, glossitis, and cervical esophageal webs. It carries an increased long term risk of squamous cell carcinoma of the esophagus and hypopharynx.

#79 · esophageal_disorders

How do peptic strictures typically appear and where are they located?

Answer

As a longer segment of smooth, tapered narrowing in the distal esophagus, caused by chronic acid-related inflammation and fibrosis. Any stricture should be biopsied to exclude malignancy.

#80 · esophageal_disorders

How is oropharyngeal dysphagia distinguished from esophageal dysphagia?

Answer

Oropharyngeal dysphagia is difficulty initiating a swallow, with coughing, choking, or nasal regurgitation right at the start of swallowing. Esophageal dysphagia is the sensation of food sticking after the swallow has already occurred.

#81 · esophageal_disorders

What is a Zenker diverticulum and how does it form?

Answer

A pharyngeal pouch that forms above the upper esophageal sphincter from increased pressure against a poorly relaxing cricopharyngeus muscle. It presents with regurgitation of old undigested food, halitosis, and aspiration risk.

#82 · esophageal_disorders

What test allows direct visualization of swallowing with different food textures by a speech pathologist?

Answer

Fiberoptic endoscopic evaluation of swallowing, which along with a modified barium swallow study helps identify the specific failing phase of swallowing and guide safe diet recommendations.

#83 · esophageal_disorders

Where in the esophagus do foreign bodies most commonly lodge?

Answer

At areas of physiologic narrowing: the upper esophageal sphincter, the level where the aortic arch crosses the esophagus, and the lower esophageal sphincter.

#84 · esophageal_disorders

Why do multiple ingested magnets require urgent removal?

Answer

They can attract to each other across the esophageal or bowel wall, causing pressure necrosis, fistula formation, or perforation, unlike a single smooth object.

#85 · esophageal_disorders

What is the appropriate timing for endoscopy in a food impaction where the patient can manage secretions?

Answer

Within twenty-four hours. Complete obstruction with inability to manage secretions is a true emergency requiring urgent same-visit endoscopic removal.

#86 · esophageal_disorders

Why should every patient with food impaction get biopsies after removal?

Answer

Because food impaction is often the first clue to an underlying condition like eosinophilic esophagitis, and biopsies are needed to look for that diagnosis.

#87 · esophageal_disorders

What is a Mallory-Weiss tear and how does it present?

Answer

A longitudinal mucosal laceration near the gastroesophageal junction from a sudden rise in intra-abdominal pressure, classically after forceful retching. It presents with hematemesis, often preceded by non-bloody vomiting.

#88 · esophageal_disorders

What medications commonly cause pill-induced esophagitis?

Answer

Doxycycline, potassium chloride, bisphosphonates, and iron supplements, especially when taken with little water or right before lying down.

#89 · esophageal_disorders

What is radiation esophagitis and what can it progress to?

Answer

Esophageal injury during or after chest radiotherapy, which can progress to fibrotic stricture over months in the previously treated field.

#90 · esophageal_disorders

What is the most common cause of esophageal perforation?

Answer

Iatrogenic injury during endoscopic procedures, followed by Boerhaave syndrome, a full thickness rupture from a sudden pressure increase against a closed glottis, usually from forceful vomiting.

#91 · esophageal_disorders

What are the classic signs of esophageal perforation?

Answer

Severe chest pain, subcutaneous emphysema, and systemic signs of sepsis as gastric contents leak into the mediastinum.

#92 · esophageal_disorders

How does scleroderma affect the esophagus?

Answer

It replaces smooth muscle in the lower two thirds with fibrous tissue, producing a weak lower esophageal sphincter and poor peristalsis, which causes severe reflux and early Barrett esophagus.

#93 · esophageal_disorders

How does Chagas disease affect the esophagus, and how does it present?

Answer

Trypanosoma cruzi destroys the myenteric plexus, producing a nonrelaxing lower esophageal sphincter and absent peristalsis identical to primary achalasia, seen in patients from endemic Latin American regions.

#94 · esophageal_disorders

How does Candida esophagitis appear on endoscopy and who gets it?

Answer

White plaques that can be scraped off the mucosa, seen in immunocompromised patients such as those with HIV, uncontrolled diabetes, or chronic steroid use, presenting with odynophagia.

#95 · esophageal_disorders

How do herpes simplex virus and cytomegalovirus esophagitis differ from Candida on endoscopy?

Answer

They cause discrete punched-out or linear ulcers rather than the white plaques seen with Candida, an important distinguishing feature in immunocompromised patients.

#96 · esophageal_disorders

What is the mechanism behind most GERD symptoms?

Answer

Transient lower esophageal sphincter relaxation that allows stomach contents to escape upward even without increased abdominal pressure, often worsened by a hiatal hernia.

#97 · esophageal_disorders

What are alarm features in a GERD patient that should prompt endoscopy rather than empiric treatment?

Answer

Dysphagia, odynophagia, weight loss, anemia, and long duration symptoms in an older patient.

#98 · esophageal_disorders

What is the gold standard test to correlate symptoms with acid or non-acid reflux?

Answer

Ambulatory pH monitoring, often combined with impedance testing, used when the diagnosis is unclear or symptoms persist despite treatment.

#99 · esophageal_disorders

Why is manometry needed before antireflux surgery?

Answer

To rule out an unrecognized motility disorder such as achalasia, which could be significantly worsened by adding a mechanical wrap around a poorly contracting esophagus.

#100 · esophageal_disorders

What extraesophageal symptoms can GERD cause?

Answer

Chronic cough, hoarseness, chest pain mimicking cardiac disease, dental erosion, and worsening of underlying asthma from chronic microaspiration.

#101 · gastric_duodenal_disorders

What are the two hallmark features of cannabinoid hyperemesis syndrome?

Answer

Recurrent cyclic vomiting in a long term heavy cannabis user, often with relief from hot showers or baths. Treatment is complete cannabis cessation, which resolves symptoms over days to weeks.

#102 · gastric_duodenal_disorders

What defines cyclic vomiting syndrome?

Answer

Discrete, stereotyped episodes of severe vomiting separated by symptom free intervals, often starting in childhood. Between episodes the patient appears entirely well, distinguishing it from a progressive structural disease.

#103 · gastric_duodenal_disorders

What triad suggests gastroparesis?

Answer

Early satiety, postprandial fullness, and vomiting of food eaten hours earlier that still looks undigested. Common causes include diabetes, prior gastric surgery, and viral illness, though many cases are idiopathic.

#104 · gastric_duodenal_disorders

What is annular pancreas?

Answer

A congenital anomaly where pancreatic tissue forms a ring around the second part of the duodenum. It can cause partial or complete duodenal obstruction in infancy or, less often, in adulthood.

#105 · gastric_duodenal_disorders

What abnormal finding on imaging suggests intestinal malrotation?

Answer

The duodenojejunal junction is abnormally positioned instead of crossing the midline near the pylorus. This finding can be seen on an upper GI contrast study in both infants and adults.

#106 · gastric_duodenal_disorders

What are the three main stimuli of parietal cell acid secretion?

Answer

Gastrin from antral G cells, histamine from enterochromaffin-like cells, and acetylcholine from vagal nerve endings. All three pathways converge on the parietal cell proton pump, the final common step.

#107 · gastric_duodenal_disorders

What is the role of somatostatin in gastric physiology?

Answer

Somatostatin is released from antral D cells in response to acid and suppresses gastrin, histamine, and acid secretion. Loss of D cell function removes this brake and can cause hypergastrinemia.

#108 · gastric_duodenal_disorders

Why does chronic PPI use raise serum gastrin?

Answer

Suppressing acid removes the normal negative feedback that acid provides on G cell gastrin release. This is a benign, expected effect and should not be mistaken for a gastrinoma.

#109 · gastric_duodenal_disorders

What is secretin's role in duodenal physiology?

Answer

Secretin is released from duodenal S cells in response to acid and stimulates pancreatic bicarbonate secretion. It also inhibits gastrin release, helping coordinate acid neutralization in the duodenum.

#110 · gastric_duodenal_disorders

What is the migrating motor complex and why does it matter?

Answer

A pattern of contractions sweeping through the stomach and small bowel between meals that clears residual debris. Failure of this pattern contributes to bezoar formation.

#111 · gastric_duodenal_disorders

What are the classic symptoms and exam finding of gastric outlet obstruction?

Answer

Early satiety, postprandial fullness, and vomiting of undigested food eaten many hours earlier. A succussion splash heard hours after eating is a classic and specific exam finding.

#112 · gastric_duodenal_disorders

What electrolyte pattern results from chronic vomiting in gastric outlet obstruction?

Answer

A hypokalemic, hypochloremic metabolic alkalosis, driven by loss of gastric acid and volume depletion causing renal potassium wasting. Fluid and electrolyte correction is a management priority.

#113 · gastric_duodenal_disorders

In an older adult with new gastric outlet obstruction and no ulcer history, what should be assumed?

Answer

Malignancy should be presumed until proven otherwise, most often gastric adenocarcinoma or pancreatic head cancer. Endoscopy with biopsy of any mass or stricture is required even if it looks benign.

#114 · gastric_duodenal_disorders

How does autoimmune gastritis differ from H pylori gastritis in distribution?

Answer

Autoimmune gastritis targets the body and fundus and spares the antrum, while H pylori gastritis typically begins in the antrum. Autoimmune gastritis leads to achlorhydria, B12 deficiency, and increased cancer risk.

#115 · gastric_duodenal_disorders

What is chemical or reactive gastropathy and what causes it?

Answer

A pattern of foveolar hyperplasia with minimal inflammation caused by bile reflux, NSAIDs, or alcohol. It is distinguished from H pylori gastritis by the lack of a lymphocytic and neutrophilic infiltrate.

#116 · gastric_duodenal_disorders

What is Menetrier disease?

Answer

A rare protein losing gastropathy with giant hypertrophied gastric folds, foveolar hyperplasia, and excess mucus causing hypoalbuminemia. It is associated with overexpression of transforming growth factor alpha.

#117 · gastric_duodenal_disorders

What is gastric antral vascular ectasia (GAVE) and what is it associated with?

Answer

A vascular lesion showing linear red stripes radiating from the pylorus, causing chronic or overt GI bleeding. It is strongly associated with systemic sclerosis and is sometimes called watermelon stomach.

#118 · gastric_duodenal_disorders

What is the cancer risk of sporadic fundic gland polyps versus those from FAP?

Answer

Sporadic fundic gland polyps, usually linked to chronic PPI use, carry essentially no cancer risk. Those arising in familial adenomatous polyposis can harbor dysplasia and need closer surveillance.

#119 · gastric_duodenal_disorders

What gastric polyp type is linked to active H pylori infection and can regress with treatment?

Answer

Hyperplastic polyps arise from chronic inflammation, often with active H pylori infection. They can regress after successful eradication therapy, though larger ones over one centimeter carry a small dysplasia risk.

#120 · gastric_duodenal_disorders

Which gastric polyp type carries the highest malignant potential?

Answer

Adenomatous polyps, though the least common type, carry the highest malignant potential. They should always be removed with attention to synchronous lesions and appropriate surveillance.

#121 · gastric_duodenal_disorders

What two mechanisms explain how NSAIDs cause ulcers?

Answer

Systemic inhibition of cyclooxygenase-1 reduces protective prostaglandin synthesis, and there is also direct topical mucosal injury. Enteric coating reduces the topical component but not the systemic risk.

#122 · gastric_duodenal_disorders

How does H pylori survive in the acidic stomach environment?

Answer

It produces urease, which breaks down urea into ammonia, neutralizing acid locally around the organism. This same urease activity is exploited by the urea breath test for diagnosis.

#123 · gastric_duodenal_disorders

Why should PPIs and antibiotics be held before H pylori testing?

Answer

They can cause false negative results on urea breath tests, stool antigen tests, and biopsy based tests. PPIs should be held one to two weeks and antibiotics or bismuth four weeks before testing.

#124 · gastric_duodenal_disorders

Why is serology not useful to confirm H pylori eradication?

Answer

Serology detects antibodies that remain positive long after successful treatment, so a positive result does not mean active infection. Urea breath test or stool antigen testing should be used instead to confirm cure.

#125 · gastric_duodenal_disorders

What endoscopic ulcer finding carries the highest rebleeding risk?

Answer

A visible vessel or active spurting in the ulcer base carries the highest rebleeding risk and warrants endoscopic therapy. A clean base ulcer carries a very low rebleeding risk and often needs no intervention.

#126 · gastric_duodenal_disorders

What is the classic finding and management of a perforated peptic ulcer?

Answer

Sudden severe abdominal pain with peritoneal signs and free air under the diaphragm on an upright chest film. This is a surgical emergency in most cases, though select stable patients may be managed nonoperatively.

#127 · gastric_duodenal_disorders

When should Zollinger-Ellison syndrome be suspected?

Answer

In patients with multiple or refractory ulcers, ulcers in unusual distal locations, or ulcers without H pylori or NSAID exposure. Roughly a quarter of cases occur with multiple endocrine neoplasia type 1.

#128 · gastric_duodenal_disorders

How is Zollinger-Ellison syndrome confirmed when gastrin is equivocal?

Answer

A secretin stimulation test is used, and a paradoxical rise in gastrin after secretin supports the diagnosis. Fasting gastrin should ideally be checked off proton pump inhibitor therapy first.

#129 · gastric_duodenal_disorders

What is early dumping syndrome and its mechanism?

Answer

Symptoms within thirty minutes of eating from rapid delivery of hyperosmolar food into the small bowel. This draws fluid into the lumen, causing cramping, diarrhea, and vasomotor flushing and tachycardia.

#130 · gastric_duodenal_disorders

What is late dumping syndrome and its mechanism?

Answer

Symptoms one to three hours after eating from an exaggerated insulin response to a rapid carbohydrate load. This causes reactive hypoglycemia with sweating, tremor, and confusion, sometimes treated with acarbose.

#131 · gastric_duodenal_disorders

What is afferent loop syndrome?

Answer

A complication of Billroth II or Roux-en-Y anatomy where the blind afferent limb carrying bile and pancreatic secretions becomes obstructed. Chronic partial obstruction causes pain relieved by sudden bilious, non-food vomiting.

#132 · gastric_duodenal_disorders

What is bile reflux gastritis and how does it present?

Answer

Reflux of duodenal contents into the gastric remnant after procedures removing or bypassing the pylorus. It causes chronic burning pain often unrelieved by eating, with bile pooling seen on endoscopy.

#133 · gastric_duodenal_disorders

What are marginal ulcers and their major risk factors?

Answer

Ulcers at the gastrojejunal anastomosis after gastric bypass, with risk factors including smoking, NSAID use, H pylori, and larger pouch size. Smoking cessation and PPI therapy are central to prevention and treatment.

#134 · gastric_duodenal_disorders

What is internal hernia after Roux-en-Y gastric bypass?

Answer

Herniation through a mesenteric defect created during reconstruction, presenting with intermittent severe pain. Imaging can be falsely reassuring, so a high index of suspicion is required.

#135 · gastric_duodenal_disorders

What nutritional deficiencies are common after gastric bypass surgery?

Answer

Iron, vitamin B12, folate, calcium, and fat soluble vitamin deficiencies are common from reduced acid processing and bypassed absorptive surface. Lifelong monitoring and supplementation are required.

#136 · gastric_duodenal_disorders

What mechanism explains diabetic gastroparesis?

Answer

Long standing hyperglycemia damages vagal nerve fibers and interstitial cells of Cajal, the gastric pacemaker cells. Poor glycemic control both results from and worsens gastroparesis in a vicious cycle.

#137 · gastric_duodenal_disorders

What must be confirmed before diagnosing gastroparesis with a gastric emptying study?

Answer

Reasonable glycemic control at the time of testing and exclusion of a mechanical obstruction on endoscopy first. Uncontrolled hyperglycemia itself can slow gastric emptying and confound the result.

#138 · gastric_duodenal_disorders

What upper GI finding can occur in Crohn disease and why does it matter?

Answer

Gastroduodenal involvement, usually as an extension of duodenal disease, can cause stricturing and gastric outlet obstruction. This should be considered in known Crohn patients with new obstructive symptoms.

#139 · gastric_duodenal_disorders

What is a gastric bezoar and what predisposes to it?

Answer

A mass of indigestible material like fiber or hair that accumulates in the stomach and can cause obstruction. Impaired gastric motility, such as from gastroparesis, predisposes by weakening the migrating motor complex.

#140 · gastric_duodenal_disorders

How does adult acquired pyloric narrowing usually differ from congenital pyloric stenosis?

Answer

Adult acquired narrowing is almost always from chronic peptic ulcer scarring rather than a true congenital muscular defect. Congenital pyloric stenosis instead involves hypertrophy of the pyloric muscle presenting in infancy.

#141 · ibd_immune_disorders

What is the depth of inflammation in Crohn disease versus ulcerative colitis?

Answer

Crohn disease is transmural, involving the full thickness of the bowel wall. Ulcerative colitis is limited to the mucosa and, at most, the submucosa. This difference explains why fistulas and strictures are common in Crohn disease but rare in ulcerative colitis.

#142 · ibd_immune_disorders

What are skip lesions and which disease shows them?

Answer

Skip lesions are patches of diseased bowel separated by areas of normal mucosa. They are a hallmark of Crohn disease. Ulcerative colitis instead spreads continuously without normal areas in between.

#143 · ibd_immune_disorders

What is the most common location for Crohn disease?

Answer

The terminal ileum is the most common site of Crohn disease involvement. Ileocolonic disease, affecting both the terminal ileum and colon, is the most frequent overall pattern seen clinically.

#144 · ibd_immune_disorders

Why does terminal ileal Crohn disease or resection cause B12 deficiency?

Answer

The terminal ileum is the primary site of vitamin B12 absorption. Disease or resection of this segment impairs B12 uptake, leading to macrocytic anemia and neurologic symptoms over time.

#145 · ibd_immune_disorders

What causes bile acid diarrhea after terminal ileal resection?

Answer

The terminal ileum reabsorbs bile acids. When it is resected or severely diseased, excess bile acids reach the colon and stimulate secretory watery diarrhea, distinct from active inflammation.

#146 · ibd_immune_disorders

What is an enterovesical fistula and how does it present?

Answer

An enterovesical fistula connects the bowel to the bladder. It classically presents with recurrent mixed enteric urinary tract infections and pneumaturia, which is air passed in the urine.

#147 · ibd_immune_disorders

What is the difference between an inflammatory and a fibrotic Crohn stricture?

Answer

An inflammatory stricture is caused by active swelling and often improves with medical anti-inflammatory therapy. A fibrotic stricture is caused by chronic scarring and generally requires endoscopic dilation or surgery instead.

#148 · ibd_immune_disorders

What imaging study is preferred to map a perianal Crohn fistula?

Answer

Pelvic MRI is the preferred imaging study for mapping perianal fistula tracts and any associated abscesses before surgical planning, since it provides detailed soft tissue resolution.

#149 · ibd_immune_disorders

Why does perianal fistulizing Crohn disease change management?

Answer

Perianal fistulizing disease marks a more aggressive overall disease course. It often prompts earlier initiation of anti-TNF biologic therapy combined with surgical drainage of any abscess.

#150 · ibd_immune_disorders

What must always be done before starting biologic therapy for a Crohn abscess?

Answer

Any abscess must be drained, either surgically or percutaneously, before or alongside starting anti-TNF therapy. Antibiotics and biologics alone will not resolve a walled-off collection.

#151 · ibd_immune_disorders

What long term cancer risk is increased in chronically diseased Crohn small bowel segments?

Answer

Chronically inflamed or surgically bypassed small bowel segments in Crohn disease carry an increased risk of small bowel adenocarcinoma, a rare but clinically important complication.

#152 · ibd_immune_disorders

How is ulcerative colitis extent classified?

Answer

Ulcerative colitis extent is classified as proctitis, limited to the rectum, left-sided colitis, extending to the splenic flexure, or extensive colitis and pancolitis, extending beyond the splenic flexure or involving the whole colon.

#153 · ibd_immune_disorders

What defines toxic megacolon?

Answer

Toxic megacolon is colonic dilation, usually greater than 6 centimeters, combined with systemic toxicity such as fever, tachycardia, leukocytosis, and abdominal distension, usually during a severe colitis flare.

#154 · ibd_immune_disorders

What medications can trigger or worsen toxic megacolon?

Answer

Opioids and anticholinergic medications slow colonic motility and are recognized triggers or aggravating factors for toxic megacolon in patients with severe colitis flares.

#155 · ibd_immune_disorders

When is urgent colectomy indicated in toxic megacolon?

Answer

Urgent colectomy is indicated if the patient fails to improve within 48 to 72 hours of intensive medical therapy or shows any sign of perforation, since delay raises mortality.

#156 · ibd_immune_disorders

Why is a colonic stricture concerning in ulcerative colitis?

Answer

Fibrotic strictures are unusual in ulcerative colitis because its inflammation is limited to the mucosa. Any stricture found should raise concern for malignancy and be evaluated before assuming benign disease.

#157 · ibd_immune_disorders

When does colorectal cancer surveillance typically begin for extensive ulcerative colitis?

Answer

Surveillance colonoscopy with extensive biopsies or chromoendoscopy typically begins about eight to ten years after diagnosis of extensive disease, assuming no primary sclerosing cholangitis is present.

#158 · ibd_immune_disorders

How does primary sclerosing cholangitis change surveillance timing?

Answer

Patients with both ulcerative colitis and primary sclerosing cholangitis need annual colonoscopic surveillance starting as soon as both diagnoses are established, much earlier than the usual eight to ten year window.

#159 · ibd_immune_disorders

What are pseudopolyps and are they premalignant?

Answer

Pseudopolyps are regenerative mucosal projections that form after healing of severe ulceration in ulcerative colitis. They are not inherently premalignant, though their presence reflects a history of severe inflammation.

#160 · ibd_immune_disorders

What is tenesmus?

Answer

Tenesmus is a persistent sensation of needing to pass stool even after passing little or nothing. It is common in ulcerative colitis, especially proctitis, because inflammation begins in the rectum.

#161 · ibd_immune_disorders

What is the difference between activity-parallel and activity-independent extraintestinal manifestations?

Answer

Activity-parallel manifestations, like peripheral arthritis and erythema nodosum, flare and improve along with bowel disease activity.
Activity-independent manifestations, like primary sclerosing cholangitis and ankylosing spondylitis, run their own separate course.

#162 · ibd_immune_disorders

Which extraintestinal skin finding is more linked to Crohn disease than ulcerative colitis?

Answer

Erythema nodosum, tender red nodules typically on the shins, is more commonly associated with Crohn disease than ulcerative colitis, although it can occur in both conditions.

#163 · ibd_immune_disorders

What does pyoderma gangrenosum look like and how does it behave?

Answer

Pyoderma gangrenosum begins as a pustule that rapidly ulcerates into a painful lesion with a violaceous, undermined border. It is activity-independent and does not reliably improve with colectomy.

#164 · ibd_immune_disorders

What liver disease is strongly associated with ulcerative colitis?

Answer

Primary sclerosing cholangitis is strongly associated with ulcerative colitis and, less commonly, Crohn colitis. It causes progressive bile duct fibrosis with a beaded appearance on MRCP and raises alkaline phosphatase.

#165 · ibd_immune_disorders

What malignancy risk does primary sclerosing cholangitis raise?

Answer

Primary sclerosing cholangitis raises the long term risk of cholangiocarcinoma, in addition to progressing toward cirrhosis, which is part of why combined surveillance with colitis is so important.

#166 · ibd_immune_disorders

What genetic markers are associated with celiac disease?

Answer

Celiac disease is strongly linked to HLA-DQ2 and HLA-DQ8, which present deamidated gluten peptides to T cells and drive the small bowel inflammatory response.

#167 · ibd_immune_disorders

What enzyme triggers the celiac immune response to gluten?

Answer

Tissue transglutaminase deamidates gluten peptides, increasing their binding to HLA-DQ2 or HLA-DQ8 on antigen presenting cells and activating the T cell response that damages the small bowel.

#168 · ibd_immune_disorders

What is the classic histologic triad of celiac disease on duodenal biopsy?

Answer

The classic celiac biopsy findings are villous atrophy, crypt hyperplasia, and increased intraepithelial lymphocytes, reflecting chronic immune-mediated damage to the small bowel lining.

#169 · ibd_immune_disorders

What is the preferred initial serologic test for celiac disease?

Answer

IgA tissue transglutaminase antibody, or tTG-IgA, is the preferred initial screening test, always checked alongside total serum IgA to rule out IgA deficiency causing a false negative.

#170 · ibd_immune_disorders

What should be done if a patient has IgA deficiency and suspected celiac disease?

Answer

If total IgA is low, tTG-IgA becomes unreliable, so IgG-based testing, such as deamidated gliadin peptide IgG, should be used instead for both diagnosis and monitoring.

#171 · ibd_immune_disorders

Why should patients keep eating gluten before celiac testing?

Answer

Starting a gluten-free diet before testing can normalize both antibody levels and villous architecture, producing false-negative serology and biopsy results and making diagnosis much harder to confirm.

#172 · ibd_immune_disorders

What is dermatitis herpetiformis?

Answer

Dermatitis herpetiformis is the classic cutaneous manifestation of celiac disease, presenting as itchy blistering lesions on the elbows, knees, and buttocks, with granular IgA deposits on skin biopsy.

#173 · ibd_immune_disorders

How is dermatitis herpetiformis treated?

Answer

Strict adherence to a gluten-free diet typically resolves dermatitis herpetiformis over time, often without needing additional topical therapy once the diet is followed consistently.

#174 · ibd_immune_disorders

Why are iron and folate deficiency classic in celiac disease?

Answer

Iron and folate are absorbed in the duodenum and proximal jejunum, the segments most damaged by celiac disease, so their deficiency can appear even before diarrhea becomes prominent.

#175 · ibd_immune_disorders

What nonclassic presentations should raise suspicion for celiac disease?

Answer

Unexplained iron deficiency anemia, elevated liver enzymes, infertility, recurrent miscarriage, short stature in children, and dental enamel defects are all recognized nonclassic celiac presentations without prominent diarrhea.

#176 · ibd_immune_disorders

What autoimmune conditions are commonly associated with celiac disease?

Answer

Autoimmune thyroid disease and type 1 diabetes are the autoimmune conditions most commonly associated with celiac disease, reflecting shared genetic and immune susceptibility.

#177 · ibd_immune_disorders

What is refractory celiac disease and why does it matter?

Answer

Refractory celiac disease is new or persistent symptoms despite strict gluten-free diet adherence. It requires repeat biopsy and workup because it can signal enteropathy-associated T-cell lymphoma.

#178 · ibd_immune_disorders

What malignancies are increased in celiac disease?

Answer

Celiac disease carries an increased risk of enteropathy-associated T-cell lymphoma and a modestly increased risk of small bowel adenocarcinoma, both linked to chronic small bowel damage.

#179 · ibd_immune_disorders

Why does untreated celiac disease cause osteoporosis?

Answer

Chronic malabsorption of calcium and vitamin D from long standing untreated celiac disease is the main mechanism leading to reduced bone density and osteoporosis over time.

#180 · ibd_immune_disorders

What is microscopic colitis and how is it diagnosed?

Answer

Microscopic colitis causes chronic watery, non-bloody diarrhea with a grossly normal-appearing colon. It is diagnosed only by biopsy, showing either a thickened collagen band or increased intraepithelial lymphocytes.

#181 · ibd_immune_disorders

What are the two subtypes of microscopic colitis?

Answer

Collagenous colitis, defined by a thickened subepithelial collagen band, and lymphocytic colitis, defined by increased intraepithelial lymphocytes, are the two histologic subtypes of microscopic colitis.

#182 · ibd_immune_disorders

Which medications are associated with microscopic colitis?

Answer

Proton pump inhibitors and nonsteroidal anti-inflammatory drugs are associated with microscopic colitis and should be reviewed and stopped if possible before starting other therapy.

#183 · ibd_immune_disorders

What is eosinophilic esophagitis and how does it present?

Answer

Eosinophilic esophagitis is dense eosinophilic infiltration of the esophagus presenting with dysphagia and food impaction in adults, diagnosed by biopsy showing a high density of eosinophils without infection.

#184 · ibd_immune_disorders

How is eosinophilic esophagitis initially managed?

Answer

Initial management includes dietary elimination of trigger foods and topical steroids that are swallowed rather than inhaled, with dilation reserved for strictures that have already formed.

#185 · ibd_immune_disorders

What is Behcet disease and how can it mimic Crohn disease?

Answer

Behcet disease is a systemic vasculitis that can cause terminal ileal and cecal ulcers mimicking Crohn disease, but with recurrent oral and genital ulcers, uveitis, and a pathergy reaction.

#186 · ibd_immune_disorders

How do Behcet disease ulcers differ from Crohn disease ulcers?

Answer

Behcet disease ulcers tend to be fewer in number but deeper and more likely to perforate compared with the more numerous, typically shallower ulcers of Crohn disease.

#187 · ibd_immune_disorders

What is a pathergy reaction?

Answer

A pathergy reaction is exaggerated skin ulceration at the site of a needle prick, a supportive clinical finding for Behcet disease.

#188 · ibd_immune_disorders

How can common variable immunodeficiency mimic inflammatory bowel disease?

Answer

Common variable immunodeficiency can cause chronic diarrhea and nodular lymphoid hyperplasia of the small bowel, overlapping with inflammatory bowel disease, but with recurrent sinopulmonary infections and low immunoglobulins.

#189 · ibd_immune_disorders

What clinical clues favor Crohn disease over ulcerative colitis on colonoscopy?

Answer

Rectal sparing, skip lesions, and terminal ileal involvement favor Crohn disease. Continuous inflammation starting in the rectum favors ulcerative colitis instead.

#190 · ibd_immune_disorders

What is the key testing rule for suspected immunodeficiency before diagnosing inflammatory bowel disease?

Answer

Recurrent infections with low immunoglobulin levels should prompt an immunodeficiency workup, such as serum immunoglobulin levels, before labeling a gastrointestinal picture as idiopathic inflammatory bowel disease.

#191 · motility_disorders

What is the underlying cell loss that causes achalasia?

Answer

Loss of inhibitory ganglion cells in the myenteric plexus of the esophagus. This causes failure of lower esophageal sphincter relaxation and loss of normal peristalsis in the esophageal body.

#192 · motility_disorders

What is the classic barium esophagram finding in achalasia?

Answer

A bird-beak appearance, with smooth tapered narrowing at the gastroesophageal junction and dilation of the esophageal body above it.

#193 · motility_disorders

What manometric finding defines achalasia on high resolution manometry?

Answer

An elevated integrated relaxation pressure combined with absent peristalsis in the esophageal body.

#194 · motility_disorders

Name the three Chicago Classification subtypes of achalasia.

Answer

Type I with no significant pressurization, type II with panesophageal pressurization, and type III with spastic contractions. Type II generally has the best treatment response.

#195 · motility_disorders

Why must pseudoachalasia be considered before treating achalasia?

Answer

A tumor at the gastroesophageal junction can produce a manometric and radiographic pattern identical to primary achalasia. Careful endoscopic and imaging evaluation is needed to exclude malignancy, especially with rapid symptom onset or significant weight loss.

#196 · motility_disorders

What limits long-term use of botulinum toxin for achalasia?

Answer

Its effect wears off over months, requiring repeated injections, which makes it best reserved for patients who are poor candidates for more durable procedures.

#197 · motility_disorders

Why is peroral endoscopic myotomy often preferred for type III achalasia?

Answer

It allows a longer myotomy that can extend up into the spastic segment of the esophageal body, unlike a standard laparoscopic Heller myotomy.

#198 · motility_disorders

What defines distal esophageal spasm on manometry?

Answer

Normal lower esophageal sphincter relaxation with premature, simultaneous contractions in the distal esophagus on multiple swallows.

#199 · motility_disorders

What defines hypercontractile (jackhammer) esophagus?

Answer

At least one swallow with a markedly elevated distal contractile integral, reflecting excessive contraction strength rather than an abnormal timing pattern.

#200 · motility_disorders

What is ineffective esophageal motility and why does it matter in reflux disease?

Answer

Weak peristalsis with low distal contractile integral on multiple swallows. It impairs esophageal clearance of refluxed acid, which can worsen gastroesophageal reflux symptoms.

#201 · motility_disorders

What is the classic symptom triad of gastroparesis?

Answer

Early satiety, postprandial fullness, and nausea, often with vomiting, bloating, and upper abdominal pain.

#202 · motility_disorders

What is the reference standard test for gastroparesis and its diagnostic threshold?

Answer

Gastric emptying scintigraphy using a low-fat egg-white meal over four hours, with retention above ten percent at four hours considered diagnostic of delayed emptying.

#203 · motility_disorders

Which two medication classes must be held before gastric emptying testing?

Answer

Opioids and glucagon-like peptide-1 receptor agonists, since both can slow gastric emptying and produce a falsely delayed result.

#204 · motility_disorders

Why is the two-hour retention value on gastric emptying scintigraphy not enough on its own?

Answer

Early time point results correlate poorly with symptoms. The four-hour retention value is the validated diagnostic threshold.

#205 · motility_disorders

What is the only prokinetic approved in the United States for gastroparesis, and its main limitation?

Answer

Metoclopramide, limited by a boxed warning for tardive dyskinesia, so treatment courses should be kept as short as possible.

#206 · motility_disorders

What options exist for refractory gastroparesis after medical therapy fails?

Answer

Gastric electrical stimulation, and for severe cases, a venting gastrostomy or jejunal feeding tube.

#207 · motility_disorders

What is the mechanism of early dumping syndrome?

Answer

Hyperosmolar chyme entering the small bowel rapidly draws fluid across the mucosa, causing bloating, cramping, diarrhea, and vasomotor symptoms within thirty minutes of eating.

#208 · motility_disorders

What is the mechanism of late dumping syndrome?

Answer

A rapid rise in blood glucose triggers an exaggerated insulin response, producing reactive hypoglycemia with sweating, tremor, and confusion one to three hours after eating.

#209 · motility_disorders

What is the most effective medical therapy for refractory dumping syndrome?

Answer

Subcutaneous octreotide, which slows gastric emptying and blunts the exaggerated insulin surge.

#210 · motility_disorders

When should prolonged postoperative ileus be investigated further?

Answer

When it persists beyond three to five days, which should prompt evaluation for electrolyte disturbance, opioid effect, retroperitoneal hematoma, or an evolving mechanical obstruction.

#211 · motility_disorders

What is acute colonic pseudo-obstruction also called, and what is its major risk?

Answer

Ogilvie syndrome. The major risk is cecal perforation, which becomes more likely once cecal diameter exceeds twelve centimeters.

#212 · motility_disorders

What is the treatment of choice for Ogilvie syndrome after conservative measures fail?

Answer

Intravenous neostigmine, given with cardiac monitoring because of the risk of bradycardia.

#213 · motility_disorders

What defines chronic intestinal pseudo-obstruction?

Answer

A rare, indolent syndrome from neuropathic or myopathic gut disease, causing recurrent episodes resembling mechanical obstruction with dilated bowel loops but no identifiable transition point.

#214 · motility_disorders

Name three underlying causes of chronic intestinal pseudo-obstruction.

Answer

Scleroderma, amyloidosis, and mitochondrial disorders are recognized primary or secondary causes, among others such as paraneoplastic syndromes.

#215 · motility_disorders

What defines dyssynergic defecation?

Answer

Failure of the puborectalis muscle and external anal sphincter to relax, or paradoxical contraction, during attempted defecation, leading to incomplete evacuation and excessive straining.

#216 · motility_disorders

What balloon expulsion time supports dyssynergic defecation?

Answer

A prolonged balloon expulsion time, typically longer than one to two minutes, when paired with an abnormal straining pattern on manometry.

#217 · motility_disorders

What is the most effective treatment for dyssynergic defecation?

Answer

Biofeedback therapy, which trains coordination of abdominal push effort with pelvic floor relaxation. It outperforms laxatives alone in randomized trials.

#218 · motility_disorders

What test identifies a structural anal sphincter defect from obstetric trauma?

Answer

Endoanal ultrasound.

#219 · motility_disorders

What is the stepwise management approach for fecal incontinence?

Answer

Stool bulking agents and antidiarrheal agents first, then biofeedback, then sacral nerve stimulation for patients who fail conservative measures.

#220 · motility_disorders

What distinguishes achalasia dysphagia from a mechanical stricture?

Answer

Achalasia causes dysphagia to both solids and liquids from the start, while a mechanical stricture or ring typically causes dysphagia to solids only, progressing later to liquids.

#221 · gut_brain_disorders

What defines functional heartburn?

Answer

Typical heartburn symptoms, a normal endoscopy, and normal acid exposure time on pH monitoring, with no meaningful correlation between symptoms and reflux events.

#222 · gut_brain_disorders

What defines reflux hypersensitivity?

Answer

Normal acid exposure time on pH monitoring but a positive symptom association with physiologic reflux episodes, meaning the esophagus is oversensitive to normal amounts of reflux.

#223 · gut_brain_disorders

How is functional chest pain diagnosed?

Answer

As a diagnosis of exclusion, after cardiac causes are ruled out and after structural and motility esophageal disease has been excluded, including normal pH testing.

#224 · gut_brain_disorders

What is globus, and does it correlate with structural disease?

Answer

A persistent sensation of a lump or tightness in the throat without true dysphagia. It does not correlate with a structural lesion once standard evaluation, including endoscopy, is unremarkable.

#225 · gut_brain_disorders

What class of medication treats functional esophageal disorders, and why?

Answer

Low-dose neuromodulators such as tricyclic antidepressants, since they act centrally to reduce visceral hypersensitivity rather than addressing acid production.

#226 · gut_brain_disorders

How is chronic nausea and vomiting syndrome distinguished from gastroparesis?

Answer

It involves bothersome chronic nausea or vomiting for at least six months with normal or only mildly delayed gastric emptying, unlike gastroparesis which requires clearly delayed emptying.

#227 · gut_brain_disorders

What defines cyclic vomiting syndrome?

Answer

Stereotyped, recurrent, discrete episodes of intense vomiting separated by symptom-free intervals, often triggered by stress or infection.

#228 · gut_brain_disorders

What condition should always be asked about in suspected cyclic vomiting syndrome, and why?

Answer

Cannabinoid hyperemesis syndrome, since it overlaps clinically with cyclic vomiting syndrome and cessation of cannabis use is curative.

#229 · gut_brain_disorders

How is functional dyspepsia defined?

Answer

Bothersome postprandial fullness, early satiety, or epigastric pain or burning, present at least three months with symptom onset at least six months earlier, and a normal upper endoscopy.

#230 · gut_brain_disorders

Name the two subtypes of functional dyspepsia.

Answer

Postprandial distress syndrome, dominated by meal-related fullness and early satiety, and epigastric pain syndrome, dominated by epigastric pain or burning not necessarily related to meals.

#231 · gut_brain_disorders

Why is Helicobacter pylori testing recommended for all dyspepsia patients?

Answer

Eradication cures a subset of patients who actually have Helicobacter pylori-related dyspepsia rather than true functional disease.

#232 · gut_brain_disorders

What is first-line therapy for epigastric pain syndrome versus postprandial distress syndrome?

Answer

Proton pump inhibitors are first-line for epigastric pain syndrome, while prokinetic agents are often favored for postprandial distress syndrome.

#233 · gut_brain_disorders

What is abdominophrenic dyssynergia?

Answer

A reflex abnormality in which the diaphragm paradoxically contracts and the abdominal wall relaxes outward, causing visible distention despite normal total gut gas volume.

#234 · gut_brain_disorders

What two conditions should be excluded in patients with unexplained bloating overlapping with irritable bowel syndrome?

Answer

Celiac disease and small intestinal bacterial overgrowth.

#235 · gut_brain_disorders

What defines centrally mediated abdominal pain syndrome?

Answer

Chronic or frequently recurring abdominal pain poorly related to eating, bowel movements, or menses, not explained by another structural or motility diagnosis, arising from altered central pain processing.

#236 · gut_brain_disorders

What defines abdominal migraine?

Answer

A pediatric-onset syndrome with paroxysmal episodes of moderate to severe periumbilical pain lasting hours, often with headache, photophobia, nausea, or vomiting, and a personal or family history of migraine.

#237 · gut_brain_disorders

What is narcotic bowel syndrome and its key clue?

Answer

Also called opioid-induced gastrointestinal hyperalgesia, it is a paradoxical worsening of chronic abdominal pain with escalating opioid doses. The key clue is pain worsening despite dose escalation.

#238 · gut_brain_disorders

What is the cornerstone of treatment for narcotic bowel syndrome?

Answer

Gradual, supervised opioid tapering combined with neuromodulator support.

#239 · gut_brain_disorders

What defines functional biliary-like pain and its management approach?

Answer

Episodic biliary-type pain without gallstones or significant structural biliary or pancreatic abnormality. Management is conservative given the modest benefit and real procedural risk of endoscopic sphincterotomy.

#240 · gut_brain_disorders

State the Rome IV frequency and duration criterion for irritable bowel syndrome.

Answer

Recurrent abdominal pain at least one day per week in the last three months, with symptom onset at least six months earlier.

#241 · gut_brain_disorders

What tool classifies irritable bowel syndrome subtype, and name the four subtypes.

Answer

The Bristol stool scale. Subtypes are irritable bowel syndrome with constipation, with diarrhea, mixed type, and unclassified.

#242 · gut_brain_disorders

What limited workup is appropriate for irritable bowel syndrome without alarm features?

Answer

Celiac serology and, in diarrhea-predominant patients, fecal calprotectin to screen for inflammatory bowel disease.

#243 · gut_brain_disorders

Name the gut-specific secretagogues used for constipation-predominant irritable bowel syndrome.

Answer

Linaclotide and lubiprostone.

#244 · gut_brain_disorders

What distinguishes levator ani syndrome from proctalgia fugax?

Answer

Levator ani syndrome causes tenderness of the puborectalis sling with pain lasting minutes to hours, while proctalgia fugax causes brief, self-limited pain lasting seconds to a few minutes with a normal exam.

#245 · gut_brain_disorders

What gut-directed psychological therapies have strong evidence in irritable bowel syndrome?

Answer

Cognitive behavioral therapy and gut-directed hypnotherapy, both of which improve global symptoms through central pain processing pathways.

#246 · neoplastic_disorders

What gene is mutated in Lynch syndrome?

Answer

A germline mutation in one of the DNA mismatch repair genes: MLH1, MSH2, MSH6, or PMS2, or an EPCAM mutation silencing MSH2. This causes microsatellite instability, screened for with tumor immunohistochemistry or PCR testing.

#247 · neoplastic_disorders

What is the most common extracolonic cancer in Lynch syndrome?

Answer

Endometrial cancer. Lynch syndrome also raises risk of ovarian, gastric, small bowel, urothelial, and biliary tract cancers.

#248 · neoplastic_disorders

What gene causes familial adenomatous polyposis (FAP)?

Answer

The APC tumor suppressor gene on chromosome 5. Loss of APC causes hundreds to thousands of colonic adenomas, with colorectal cancer nearly universal by age 40 without colectomy.

#249 · neoplastic_disorders

What is Gardner syndrome?

Answer

An FAP variant that adds osteomas, dental abnormalities, epidermoid cysts, and desmoid tumors to the classic colonic polyposis picture.

#250 · neoplastic_disorders

What is Turcot syndrome?

Answer

FAP or Lynch syndrome combined with a central nervous system tumor. Medulloblastoma occurs with FAP, and glioblastoma occurs with Lynch syndrome.

#251 · neoplastic_disorders

What system stages duodenal and periampullary adenomas in FAP?

Answer

The Spigelman staging system, which accounts for polyp number, size, histology, and dysplasia grade to guide surveillance intervals.

#252 · neoplastic_disorders

What gene causes familial atypical multiple mole melanoma (FAMMM) syndrome?

Answer

CDKN2A, which encodes the tumor suppressor p16. This syndrome combines melanoma risk with a markedly increased lifetime pancreatic cancer risk, near 15 to 20 percent.

#253 · neoplastic_disorders

What gene causes Peutz-Jeghers syndrome?

Answer

STK11. It produces mucocutaneous pigmented macules on the lips and buccal mucosa plus hamartomatous GI polyps that can cause recurrent intussusception.

#254 · neoplastic_disorders

Which BRCA gene carries a stronger pancreatic cancer association?

Answer

BRCA2. BRCA2-associated pancreatic cancers may respond well to platinum chemotherapy and PARP inhibitors.

#255 · neoplastic_disorders

What genes cause juvenile polyposis syndrome, and what other condition is linked?

Answer

SMAD4 or BMPR1A. SMAD4 mutations also link to hereditary hemorrhagic telangiectasia, requiring screening for arteriovenous malformations.

#256 · neoplastic_disorders

What is the most common benign esophageal tumor?

Answer

Leiomyoma, arising from smooth muscle in the muscularis propria, appearing as a hypoechoic submucosal lesion on endoscopic ultrasound.

#257 · neoplastic_disorders

What are the main risk factors for esophageal squamous cell carcinoma?

Answer

Tobacco and alcohol use acting synergistically, plus very hot beverages, achalasia, caustic strictures, and Plummer-Vinson syndrome.

#258 · neoplastic_disorders

What is the dominant risk factor for esophageal adenocarcinoma?

Answer

Chronic gastroesophageal reflux disease leading to Barrett esophagus, intestinal metaplasia of the distal esophagus. Obesity is an independent risk factor.

#259 · neoplastic_disorders

How often is nondysplastic Barrett esophagus monitored?

Answer

Surveillance endoscopy every 3 to 5 years.

#260 · neoplastic_disorders

How is confirmed high-grade dysplasia in Barrett esophagus treated?

Answer

Endoscopic eradication therapy, usually radiofrequency ablation combined with resection of any visible nodularity.

#261 · neoplastic_disorders

What imaging modality is primary for local T and N staging of esophageal cancer?

Answer

Endoscopic ultrasound, combined with CT and PET for distant metastasis assessment.

#262 · neoplastic_disorders

What gastric polyp type is linked to chronic proton pump inhibitor use?

Answer

Fundic gland polyps, small and multiple in the gastric body and fundus, with essentially no malignant potential when sporadic.

#263 · neoplastic_disorders

What causes gastric hyperplastic polyps and how do they respond to treatment?

Answer

Chronic gastritis, often from H. pylori infection. They can regress after H. pylori eradication; those over 1 cm should be resected.

#264 · neoplastic_disorders

What are the two Lauren classification subtypes of gastric adenocarcinoma?

Answer

Intestinal type, forming gland-like structures and linked to H. pylori and atrophic gastritis, and diffuse type, with signet ring cells causing linitis plastica and linked to CDH1 mutations.

#265 · neoplastic_disorders

What is linitis plastica?

Answer

A rigid, thickened stomach wall caused by diffuse infiltration of poorly cohesive signet ring cells, the hallmark of diffuse type gastric adenocarcinoma.

#266 · neoplastic_disorders

What treatment can cure early localized gastric MALT lymphoma?

Answer

H. pylori eradication therapy alone, since early localized MALT lymphoma frequently regresses completely once the infection is cleared.

#267 · neoplastic_disorders

What immunohistochemical marker confirms a gastrointestinal stromal tumor (GIST)?

Answer

CD117 (KIT). Most GISTs carry activating KIT or PDGFRA mutations and arise from the interstitial cells of Cajal.

#268 · neoplastic_disorders

What drug treats unresectable or metastatic GIST?

Answer

Imatinib, a tyrosine kinase inhibitor targeting the KIT or PDGFRA mutation driving the tumor.

#269 · neoplastic_disorders

Where does small bowel adenocarcinoma most often arise?

Answer

The duodenum, particularly near the ampulla. It shares risk factors with colorectal cancer including Lynch syndrome, FAP, and Crohn disease.

#270 · neoplastic_disorders

When does carcinoid syndrome typically develop in small bowel neuroendocrine tumors?

Answer

Only once liver metastases are present, since the liver normally metabolizes vasoactive substances like serotonin before they reach systemic circulation.

#271 · neoplastic_disorders

What small bowel lymphoma is linked to poorly controlled celiac disease?

Answer

Enteropathy-associated T cell lymphoma.

#272 · neoplastic_disorders

What is the most common benign liver tumor and its imaging pattern?

Answer

Hepatic hemangioma, showing peripheral nodular enhancement with progressive fill-in on contrast imaging. Biopsy is avoided due to bleeding risk.

#273 · neoplastic_disorders

What is the classic imaging finding of focal nodular hyperplasia?

Answer

A central scar with a spoke-wheel pattern of arterial supply. It carries no malignant potential and is simply observed.

#274 · neoplastic_disorders

What is hepatic adenoma strongly linked to, and what is its main risk?

Answer

Oral contraceptive and anabolic steroid use. It carries real hemorrhage risk, especially over 5 cm, and a smaller malignant transformation risk.

#275 · neoplastic_disorders

Why is hepatitis B unusual among causes of hepatocellular carcinoma?

Answer

It can cause hepatocellular carcinoma even without cirrhosis, through direct viral integration into host DNA.

#276 · neoplastic_disorders

What is the hepatocellular carcinoma surveillance protocol in cirrhotic patients?

Answer

Ultrasound, with or without alpha-fetoprotein, every 6 months.

#277 · neoplastic_disorders

How can hepatocellular carcinoma be diagnosed without biopsy in cirrhotic patients?

Answer

A nodule showing arterial phase hyperenhancement with portal venous or delayed phase washout meets standardized imaging criteria for diagnosis.

#278 · neoplastic_disorders

What is the second most common primary liver cancer?

Answer

Intrahepatic cholangiocarcinoma, with risk factors including primary sclerosing cholangitis, liver fluke infection, and hepatolithiasis.

#279 · neoplastic_disorders

Why are choledochal cysts surgically excised?

Answer

They carry a significantly increased lifetime risk of cholangiocarcinoma despite being congenital malformations rather than true neoplasms.

#280 · neoplastic_disorders

What gallbladder polyp features suggest a true neoplasm rather than a cholesterol polyp?

Answer

Size over 1 cm, growth on serial imaging, a sessile shape, and coexisting gallstones or primary sclerosing cholangitis.

#281 · neoplastic_disorders

What is a porcelain gallbladder, and what cancer is it linked to?

Answer

Extensive dystrophic calcification of the gallbladder wall, strongly associated with gallbladder carcinoma, especially with large gallstones.

#282 · neoplastic_disorders

What is a Klatskin tumor?

Answer

A perihilar cholangiocarcinoma occurring at the confluence of the right and left hepatic ducts.

#283 · neoplastic_disorders

What is the strongest Western risk factor for cholangiocarcinoma?

Answer

Primary sclerosing cholangitis, especially with coexisting inflammatory bowel disease.

#284 · neoplastic_disorders

Which pancreatic cyst has essentially no malignant potential?

Answer

Serous cystadenoma, showing a microcystic honeycomb appearance with a central scar.

#285 · neoplastic_disorders

How does mucinous cystic neoplasm differ from IPMN?

Answer

Mucinous cystic neoplasm occurs almost exclusively in women, usually body or tail, and does not communicate with the main duct, unlike IPMN which does.

#286 · neoplastic_disorders

Which IPMN feature carries the highest malignant risk?

Answer

Main pancreatic duct involvement, along with worrisome features like an enhancing mural nodule or dilated main duct.

#287 · neoplastic_disorders

What syndrome does gastrinoma cause, and what are its features?

Answer

Zollinger-Ellison syndrome, with refractory peptic ulcers and diarrhea from excess gastrin-driven acid secretion.

#288 · neoplastic_disorders

What is Courvoisier sign?

Answer

A palpable, nontender, distended gallbladder from distal biliary obstruction, classically seen with pancreatic head cancer causing painless jaundice.

#289 · neoplastic_disorders

What limits CA 19-9 as a pancreatic cancer marker?

Answer

It can be falsely low in patients lacking the Lewis antigen, and it lacks specificity since benign biliary obstruction can also elevate it.

#290 · neoplastic_disorders

What is the adenoma to carcinoma sequence?

Answer

Stepwise mutation accumulation: APC loss allowing proliferation, then KRAS mutation, then loss of 18q and DCC/SMAD4, then TP53 loss enabling invasion.

#291 · neoplastic_disorders

Why are sessile serrated lesions easy to miss on colonoscopy?

Answer

They are flat, often right-sided, and covered with a mucous cap, contributing to interval right-sided cancers despite recent clean colonoscopies.

#292 · neoplastic_disorders

At what age does average risk colorectal cancer screening begin?

Answer

Age 45, using colonoscopy every 10 years or a stool-based test at its own recommended interval.

#293 · neoplastic_disorders

What tumor molecular findings predict poor response to anti-EGFR therapy in colorectal cancer?

Answer

RAS mutations, including KRAS, which predict lack of response to anti-EGFR antibody therapy.

#294 · neoplastic_disorders

What molecular finding predicts good response to checkpoint inhibitors in colorectal cancer?

Answer

Microsatellite instability-high status, often related to Lynch syndrome or sporadic mismatch repair deficiency.

#295 · neoplastic_disorders

What is a Sister Mary Joseph nodule?

Answer

A palpable periumbilical metastatic nodule, most often from gastric, pancreatic, ovarian, or colonic adenocarcinoma, usually signaling widespread metastatic disease.

#296 · liver_disorders

What percentage of hepatic blood flow comes from the portal vein versus the hepatic artery?

Answer

The portal vein supplies about 70 to 75 percent of total hepatic blood flow, while the hepatic artery supplies the remaining 25 to 30 percent. This dual supply explains why isolated hepatic artery occlusion rarely causes infarction in a normal liver.

#297 · liver_disorders

What is the Couinaud classification system?

Answer

It divides the liver into eight functional segments based on portal vein branching and hepatic vein drainage rather than the older right and left lobe anatomy. Each segment has independent vascular supply and outflow, allowing surgical resection of individual segments.

#298 · liver_disorders

Why does INR rise before albumin falls in acute liver injury?

Answer

Clotting factors have much shorter half-lives, hours to a couple of days, than albumin, which has a half-life of about three weeks. A drop in hepatic synthesis therefore shows up in the INR much faster than in serum albumin.

#299 · liver_disorders

What does isolated unconjugated hyperbilirubinemia suggest?

Answer

It suggests either increased bilirubin production from hemolysis or impaired hepatocyte conjugation, as in Gilbert syndrome. This differs from conjugated hyperbilirubinemia, which points to hepatocellular injury or biliary obstruction.

#300 · liver_disorders

What is enterohepatic circulation of bile acids?

Answer

Bile acids made from cholesterol are secreted into bile, delivered to the intestine, and reabsorbed in the terminal ileum to be recycled back to the liver. Disruption from ileal disease or resection reduces the bile acid pool and impairs fat absorption.

#301 · liver_disorders

What does an AST to ALT ratio greater than 2 suggest?

Answer

It suggests alcohol-related liver disease. Chronic alcohol use depletes pyridoxal phosphate, a cofactor needed more heavily for ALT synthesis than AST, producing a disproportionately low ALT relative to AST.

#302 · liver_disorders

What is the cholestatic biochemical pattern of liver injury?

Answer

Elevated alkaline phosphatase and gamma-glutamyl transferase with relatively normal aminotransferases, reflecting injury to the biliary system rather than hepatocytes. This contrasts with the hepatocellular pattern of elevated AST and ALT.

#303 · liver_disorders

What laboratory values make up the MELD-Na score?

Answer

Creatinine, bilirubin, INR, and sodium. It predicts 90-day mortality and is used to prioritize patients for liver transplant allocation.

#304 · liver_disorders

What laboratory and clinical values make up the Child-Pugh score?

Answer

Bilirubin, albumin, INR, presence and severity of ascites, and grade of hepatic encephalopathy. It classifies cirrhosis severity into classes A, B, and C and is useful for estimating surgical risk.

#305 · liver_disorders

Why is transjugular liver biopsy sometimes preferred over percutaneous biopsy?

Answer

It is safer in patients with coagulopathy or significant ascites, and it allows simultaneous measurement of the hepatic venous pressure gradient during the same procedure.

#306 · liver_disorders

What does vibration-controlled transient elastography measure?

Answer

It measures shear wave velocity through liver tissue, which correlates with liver stiffness and therefore fibrosis stage. Faster velocity indicates stiffer, more fibrotic tissue.

#307 · liver_disorders

What advantage does magnetic resonance elastography have over transient elastography?

Answer

It assesses the entire liver rather than a small sampled region, improving accuracy in patients with heterogeneous fibrosis or obesity, though it is more expensive and less available.

#308 · liver_disorders

What is the FIB-4 index used for?

Answer

It is an inexpensive noninvasive fibrosis screening calculator using age, AST, ALT, and platelet count to identify patients who need further fibrosis assessment such as elastography or biopsy.

#309 · liver_disorders

How is the hepatic venous pressure gradient calculated and what is a normal value?

Answer

It is the wedged hepatic venous pressure minus the free hepatic venous pressure. A normal gradient is 1 to 5 mmHg.

#310 · liver_disorders

What hepatic venous pressure gradient defines clinically significant portal hypertension?

Answer

A gradient of 10 mmHg or higher defines clinically significant portal hypertension, the threshold above which varices, ascites, and decompensation become likely.

#311 · liver_disorders

What hepatic venous pressure gradient threshold is most associated with variceal bleeding risk?

Answer

A gradient of 12 mmHg or higher is the threshold most closely associated with the risk of variceal hemorrhage.

#312 · liver_disorders

What are the three anatomic categories of portal hypertension resistance?

Answer

Presinusoidal, such as portal vein thrombosis, sinusoidal, dominated by cirrhosis, and postsinusoidal, such as hepatic vein outflow obstruction or severe right heart failure.

#313 · liver_disorders

What drives the hyperdynamic circulatory state in decompensated cirrhosis?

Answer

Excess nitric oxide causes splanchnic arterial vasodilation, increasing portal venous inflow and worsening the hemodynamic derangement of portal hypertension over time.

#314 · liver_disorders

What four major complications does portal hypertension directly cause?

Answer

Varices, ascites and hepatic hydrothorax, hepatorenal syndrome, and hepatic encephalopathy. Portal hypertension is the shared upstream driver of all four.

#315 · liver_disorders

What is the first-line pharmacologic therapy for primary prophylaxis of a first variceal bleed?

Answer

Nonselective beta blockers such as propranolol, nadolol, or carvedilol, which reduce portal pressure through decreased cardiac output and splanchnic vasoconstriction.

#316 · liver_disorders

What endoscopic finding on varices predicts a higher risk of bleeding?

Answer

Red wale marks, longitudinal red streaks on the variceal surface indicating a thinned wall, along with larger variceal size.

#317 · liver_disorders

Why are prophylactic antibiotics given during acute variceal hemorrhage?

Answer

Bacterial infection is common in cirrhotic patients with variceal bleeding and worsens rebleeding risk and mortality, so antibiotics are given regardless of documented infection.

#318 · liver_disorders

Why is overtransfusion avoided during acute variceal hemorrhage?

Answer

Overtransfusion to a high hemoglobin target can raise portal pressure and provoke rebleeding, so a restrictive transfusion strategy is preferred.

#319 · liver_disorders

What is the preferred endoscopic treatment for bleeding gastric varices?

Answer

Injection of a tissue adhesive such as cyanoacrylate, since band ligation is technically difficult and less effective in the thicker-walled gastric fundus.

#320 · liver_disorders

What is TIPS and what is its main drawback?

Answer

A transjugular intrahepatic portosystemic shunt directly connects the portal and hepatic veins to decompress portal pressure rapidly. It can precipitate or worsen hepatic encephalopathy because it diverts unfiltered portal blood into systemic circulation.

#321 · liver_disorders

What is balloon tamponade used for in variceal bleeding?

Answer

It is a temporary bridge to definitive therapy such as TIPS in massive hemorrhage refractory to other measures, not a definitive treatment on its own.

#322 · liver_disorders

What is the serum-ascites albumin gradient and its diagnostic cutoff?

Answer

Serum albumin minus ascitic fluid albumin. A SAAG of 1.1 g/dL or higher indicates portal hypertension as the cause with about 97 percent sensitivity, while a lower value suggests a peritoneal cause.

#323 · liver_disorders

What ascitic fluid neutrophil count diagnoses spontaneous bacterial peritonitis?

Answer

An absolute neutrophil count of 250 cells per cubic millimeter or higher without an evident surgical source of infection.

#324 · liver_disorders

When is intravenous albumin given for SBP and why?

Answer

On day 1 and day 3 of treatment, which reduces the risk of hepatorenal syndrome in patients with significant renal impairment or elevated bilirubin.

#325 · liver_disorders

What is the standard first-line diuretic regimen for cirrhotic ascites?

Answer

Combined spironolactone and furosemide, typically in a ratio of about 100 to 40 mg, titrated together with sodium restriction to maintain potassium balance.

#326 · liver_disorders

How is refractory ascites defined?

Answer

Ascites that does not respond to maximal tolerated diuretic doses with sodium restriction, or that recurs rapidly after large volume paracentesis.

#327 · liver_disorders

What is hepatic hydrothorax and why does a chest tube get avoided?

Answer

A pleural effusion, usually right sided, from ascitic fluid passing through small diaphragmatic defects into the pleural space. Chest tubes cause large ongoing protein and fluid losses without treating the underlying portal hypertension.

#328 · liver_disorders

What is hepatorenal syndrome at its core?

Answer

A functional, hemodynamically mediated form of acute kidney injury in advanced cirrhosis with ascites, caused by intense renal vasoconstriction from effective arterial underfilling, without intrinsic structural kidney disease.

#329 · liver_disorders

What must be excluded before diagnosing hepatorenal syndrome?

Answer

Volume-responsive prerenal azotemia, shock, recent nephrotoxic drug exposure, and structural kidney disease, confirmed by failure to improve after two days of diuretic withdrawal and albumin volume expansion.

#330 · liver_disorders

What is the preferred medical treatment for hepatorenal syndrome?

Answer

Terlipressin plus albumin where available. In the United States, midodrine, octreotide, and albumin combination therapy is commonly used instead.

#331 · liver_disorders

What is the only definitive cure for hepatorenal syndrome?

Answer

Liver transplantation, since it is the only treatment that reverses the underlying splanchnic vasodilation and portal hypertension driving renal vasoconstriction.

#332 · liver_disorders

What causes ammonia to accumulate and cause hepatic encephalopathy?

Answer

Reduced hepatocyte urea cycle function or portosystemic shunting allows gut-derived ammonia to bypass hepatic clearance, accumulate in the blood, and cross into the brain.

#333 · liver_disorders

What happens to ammonia once it crosses the blood brain barrier?

Answer

Astrocytes convert it to glutamine, and the resulting osmotic buildup causes astrocyte swelling that disrupts normal neurotransmission.

#334 · liver_disorders

What is the West Haven grading system used for?

Answer

It grades hepatic encephalopathy severity from grade 0, minimal changes detectable only on specialized testing, to grade 4, coma.

#335 · liver_disorders

Is asterixis specific for hepatic encephalopathy?

Answer

No. Asterixis is a classic finding in grade 2 encephalopathy and above but is nonspecific, occurring in other metabolic encephalopathies such as uremia and hypercapnia.

#336 · liver_disorders

Name five common precipitants of hepatic encephalopathy.

Answer

Gastrointestinal bleeding, infection, constipation, dehydration or diuretic overuse, and electrolyte disturbances such as hypokalemia. Sedative use and new portosystemic shunts including TIPS are also common triggers.

#337 · liver_disorders

How does lactulose treat hepatic encephalopathy?

Answer

Colonic bacteria metabolize it into acidic byproducts that trap ammonia as ammonium in the gut lumen, and its laxative effect reduces colonic transit time and toxin absorption.

#338 · liver_disorders

When is rifaximin added to encephalopathy treatment?

Answer

For patients with recurrent encephalopathy episodes despite adequate lactulose dosing. It is a poorly absorbed antibiotic that reduces ammoniagenic gut bacteria.

#339 · liver_disorders

Why is routine protein restriction no longer recommended for hepatic encephalopathy?

Answer

It worsens malnutrition and sarcopenia, and adequate protein supports muscle mass, which helps clear ammonia when hepatic clearance is impaired.

#340 · liver_disorders

What is hepatopulmonary syndrome and what are its classic symptoms?

Answer

Intrapulmonary vascular dilation causing ventilation-perfusion mismatch in liver disease and portal hypertension. Classic symptoms are platypnea, dyspnea worsened when upright, and orthodeoxia, oxygen desaturation when upright.

#341 · liver_disorders

How is hepatopulmonary syndrome confirmed?

Answer

Contrast-enhanced echocardiography with agitated saline showing delayed appearance of microbubbles in the left heart, confirming intrapulmonary shunting.

#342 · liver_disorders

What distinguishes portopulmonary hypertension from hepatopulmonary syndrome?

Answer

Portopulmonary hypertension involves pulmonary arterial vasoconstriction and remodeling with elevated mean pulmonary artery pressure and pulmonary vascular resistance but normal wedge pressure, unlike the vascular dilation seen in hepatopulmonary syndrome.

#343 · liver_disorders

What is the classic triad of Budd-Chiari syndrome and its cause?

Answer

Abdominal pain, ascites, and hepatomegaly, caused by hepatic venous outflow obstruction, usually from an underlying thrombotic tendency. The caudate lobe is often spared due to separate venous drainage into the inferior vena cava.

#344 · liver_disorders

What causes periportal pipestem fibrosis and portal hypertension in schistosomiasis?

Answer

Egg deposition and granuloma formation in small portal venules causes presinusoidal fibrosis that obstructs portal flow while largely sparing hepatocyte function, giving a comparatively favorable prognosis if treated.

#345 · liver_disorders

What condition should be suspected when a patient has severe portal hypertension complications but normal bilirubin, albumin, and INR?

Answer

A noncirrhotic cause of portal hypertension, such as idiopathic noncirrhotic portal hypertension or congenital hepatic fibrosis, especially if the patient has coexisting polycystic kidney disease.

#346 · liver_disorders

What is the key serologic difference between vaccination and resolved natural hepatitis B infection?

Answer

Vaccination produces only a positive anti-HBs with everything else negative, since the vaccine contains no core antigen. Resolved natural infection shows a positive total anti-HBc along with the positive anti-HBs, reflecting real past exposure to the whole virus.

#347 · liver_disorders

What does a positive IgM anti-HBc specifically indicate?

Answer

Recent, acute hepatitis B infection within roughly the past six months. It is negative in chronic infection, resolved infection, and vaccination, making it the key marker that isolates a truly acute case.

#348 · liver_disorders

Why is hepatitis D infection always seen only with hepatitis B?

Answer

Hepatitis D is a defective virus that cannot make its own envelope and must borrow hepatitis B surface antigen to form complete viral particles. Without hepatitis B infection, hepatitis D cannot establish infection at all.

#349 · liver_disorders

Coinfection versus superinfection in hepatitis D, what is the difference?

Answer

Coinfection is simultaneous acquisition of hepatitis B and D together, which often resolves together. Superinfection is hepatitis D infecting someone with existing chronic hepatitis B, which typically causes more severe and rapidly progressive liver disease.

#350 · liver_disorders

What confirms active hepatitis C infection after a positive antibody screen?

Answer

HCV RNA testing. A positive antibody alone only shows past exposure, which may have already cleared, so RNA testing is required before considering treatment for active infection.

#351 · liver_disorders

Does curing hepatitis C with direct-acting antivirals eliminate liver cancer risk in cirrhosis?

Answer

No. Sustained virologic response reduces but does not eliminate hepatocellular carcinoma risk once cirrhosis is present, so surveillance with ultrasound and alpha-fetoprotein must continue indefinitely.

#352 · liver_disorders

Which autoantibodies characterize type 1 versus type 2 autoimmune hepatitis?

Answer

Type 1, more common in adults, is associated with ANA and anti-smooth muscle antibody. Type 2, more common in children, is associated with anti-liver kidney microsomal antibody (anti-LKM1).

#353 · liver_disorders

What is the classic biopsy finding in autoimmune hepatitis?

Answer

Interface hepatitis with a plasma-cell-rich inflammatory infiltrate. This distinguishes it from the bile-duct-centered lymphocytic injury seen in primary biliary cholangitis.

#354 · liver_disorders

What is standard induction therapy for autoimmune hepatitis?

Answer

Corticosteroids combined with azathioprine as a steroid-sparing maintenance agent. Many patients need long-term low-dose therapy since relapse is common if treatment is withdrawn too early.

#355 · liver_disorders

What antibody is positive in the large majority of primary biliary cholangitis patients?

Answer

Antimitochondrial antibody. Its presence, together with a cholestatic enzyme pattern and typical demographic, usually makes liver biopsy unnecessary to confirm the diagnosis.

#356 · liver_disorders

What is first-line therapy for primary biliary cholangitis, and what if response is inadequate?

Answer

Ursodeoxycholic acid is first-line. If the biochemical response is inadequate after one year, obeticholic acid or a fibrate can be added as second-line therapy.

#357 · liver_disorders

What symptom often precedes jaundice by years in primary biliary cholangitis?

Answer

Pruritus, often accompanied by fatigue, frequently appears well before any visible jaundice and can severely affect quality of life even in early disease.

#358 · liver_disorders

What is the classic cholangiographic finding in primary sclerosing cholangitis?

Answer

A beaded appearance of multifocal strictures and dilations affecting both intrahepatic and extrahepatic bile ducts, reflecting chronic inflammation and fibrosis of the biliary tree.

#359 · liver_disorders

What inflammatory bowel disease is most strongly linked to primary sclerosing cholangitis?

Answer

Ulcerative colitis. Patients with both conditions need earlier and more frequent colorectal cancer surveillance than patients with ulcerative colitis alone.

#360 · liver_disorders

How does IgG4-related sclerosing cholangitis differ from primary sclerosing cholangitis?

Answer

IgG4-related disease responds well to corticosteroids and is not linked to ulcerative colitis, while primary sclerosing cholangitis does not respond to steroids and has a strong ulcerative colitis association.

#361 · liver_disorders

What cancer must be excluded with any new dominant stricture in primary sclerosing cholangitis?

Answer

Cholangiocarcinoma. Patients with primary sclerosing cholangitis carry a significantly increased lifetime risk, so a new stricture cannot be assumed benign without evaluation.

#362 · liver_disorders

What AST to ALT ratio is classic for alcohol-associated liver injury?

Answer

Greater than two. This pattern, along with transaminases rarely rising above the low hundreds, helps distinguish alcohol-related injury from viral or metabolic causes of hepatitis.

#363 · liver_disorders

What score identifies severe alcoholic hepatitis warranting corticosteroid consideration?

Answer

The Maddrey discriminant function. A score of 32 or higher defines severe disease, for which a course of corticosteroids can be considered absent contraindications like active infection.

#364 · liver_disorders

What is the single most important intervention across every stage of alcohol-associated liver disease?

Answer

Sustained abstinence from alcohol. It can even reverse simple steatosis and remains the most important factor in outcomes regardless of disease stage.

#365 · liver_disorders

What noninvasive score is the first step to risk-stratify fibrosis in suspected MASLD?

Answer

The FIB-4 index, which combines age, transaminases, and platelet count. An elevated result prompts further testing such as transient elastography before considering biopsy.

#366 · liver_disorders

What weight loss target improves both steatosis and fibrosis in MASH?

Answer

At least seven to ten percent body weight loss achieved through diet and exercise, which has demonstrated benefit for both the fat content and fibrosis stage of the liver.

#367 · liver_disorders

What is the leading cause of death in most patients with MASLD?

Answer

Cardiovascular disease, not liver-related complications, reflecting the shared metabolic risk factors between fatty liver disease and cardiovascular disease.

#368 · liver_disorders

What genotype of alpha-1 antitrypsin deficiency carries the highest liver disease risk?

Answer

PiZZ. The abnormal Z-variant protein is not properly secreted and accumulates within hepatocytes, causing direct injury, and this genotype also carries the highest emphysema risk.

#369 · liver_disorders

Why is liver transplant curative for alpha-1 antitrypsin deficiency liver disease?

Answer

Because the new liver produces normal protein instead of the misfolded variant, eliminating the hepatocyte injury mechanism, even though the lung disease risk from the original genotype remains.

#370 · liver_disorders

What gene is mutated in the majority of hereditary hemochromatosis cases?

Answer

HFE. This mutation leads to excessive intestinal iron absorption, eventually causing iron deposition in the liver, heart, pancreas, joints, and skin if untreated.

#371 · liver_disorders

What is first-line treatment for hereditary hemochromatosis?

Answer

Therapeutic phlebotomy, started as early as possible once diagnosis is confirmed, to prevent irreversible organ damage even in patients without symptoms yet.

#372 · liver_disorders

What gene is mutated in Wilson disease, and what does it normally do?

Answer

ATP7B, which normally helps excrete copper into bile. Its mutation causes copper accumulation in the liver, brain, and eyes, producing combined liver and neuropsychiatric disease.

#373 · liver_disorders

What eye finding is classic for Wilson disease?

Answer

Kayser-Fleischer rings, seen on slit-lamp examination, caused by copper deposition in the cornea. They occur alongside low ceruloplasmin and elevated urinary copper excretion.

#374 · liver_disorders

What liver disease mechanism is seen in cystic fibrosis?

Answer

Thick, obstructive biliary secretions from abnormal CFTR function, which can progress from focal biliary cirrhosis to overt cirrhosis with portal hypertension in some patients.

#375 · liver_disorders

What is sickle cell hepatopathy, and how is severe intrahepatic cholestasis treated?

Answer

A spectrum of sickle cell disease-related liver injury from sinusoidal obstruction by sickled cells. Severe intrahepatic cholestasis is treated with exchange transfusion.

#376 · liver_disorders

How does progressive familial intrahepatic cholestasis differ from benign recurrent intrahepatic cholestasis?

Answer

Both involve the same bile transport genes, but progressive familial disease causes chronic progressive cholestasis, while benign recurrent disease causes self-limited episodes that fully resolve between attacks.

#377 · liver_disorders

What tool guides N-acetylcysteine treatment after a single acute acetaminophen overdose?

Answer

The Rumack-Matthew nomogram, which plots serum acetaminophen level against time since ingestion to determine whether treatment is indicated.

#378 · liver_disorders

What is the difference between predictable and idiosyncratic drug-induced liver injury?

Answer

Predictable injury, like acetaminophen toxicity, is dose-dependent and expected. Idiosyncratic injury, such as from amoxicillin-clavulanate, is unpredictable and not clearly related to dose.

#379 · liver_disorders

What classic triad suggests Budd-Chiari syndrome?

Answer

Abdominal pain, ascites, and hepatomegaly from hepatic venous outflow obstruction, often driven by an underlying myeloproliferative neoplasm or other hypercoagulable state.

#380 · liver_disorders

What is sinusoidal obstruction syndrome, and when does it classically occur?

Answer

Previously called veno-occlusive disease, it presents with painful hepatomegaly, weight gain, and jaundice, classically within weeks of hematopoietic stem cell transplantation or certain chemotherapy.

#381 · liver_disorders

What pregnancy-related liver condition presents with pruritus of the palms and soles?

Answer

Intrahepatic cholestasis of pregnancy, which carries a real stillbirth risk and is treated with ursodeoxycholic acid along with consideration of early delivery.

#382 · liver_disorders

What is the definitive treatment for HELLP syndrome and acute fatty liver of pregnancy?

Answer

Prompt delivery. Both conditions resolve once the pregnancy ends, and continuing the pregnancy risks serious maternal and fetal complications.

#383 · liver_disorders

What defines acute liver failure in a patient without preexisting cirrhosis?

Answer

Coagulopathy with INR at or above 1.5 plus any degree of encephalopathy, developing within 26 weeks of illness onset in someone without prior chronic liver disease.

#384 · liver_disorders

What is the most feared complication of acute liver failure with high-grade encephalopathy?

Answer

Cerebral edema, a leading cause of death that requires close monitoring of intracranial pressure in patients with grade 3 or 4 encephalopathy.

#385 · liver_disorders

What defines acute-on-chronic liver failure, and what are its common precipitants?

Answer

Acute decompensation in a patient with known chronic liver disease plus failure of one or more extrahepatic organs. Common precipitants include infection, gastrointestinal bleeding, and alcohol relapse.

#386 · liver_disorders

What must be avoided when evaluating a suspected echinococcal (hydatid) liver cyst?

Answer

Percutaneous aspiration without proper precautions, since spillage of cyst contents can cause severe anaphylaxis. Treatment combines albendazole with careful drainage such as the PAIR technique.

#387 · liver_disorders

What causes congestive hepatopathy, and what can it progress to if untreated?

Answer

Sinusoidal congestion from elevated central venous pressure, usually from right heart failure or constrictive pericarditis, which can progress to cardiac cirrhosis if left untreated.

#388 · liver_disorders

What score is used to prioritize patients for liver transplant, and what does it predict?

Answer

The MELD-Na score, which predicts short-term mortality in decompensated cirrhosis and prioritizes the sickest patients for organ allocation on the waiting list.

#389 · liver_disorders

What class of medication forms the backbone of long-term post-transplant immunosuppression?

Answer

Calcineurin inhibitors such as tacrolimus, balanced against risks of infection, malignancy, chronic kidney disease, and recurrence of the original liver disease.

#390 · liver_disorders

Which liver diseases are most likely to recur after transplantation?

Answer

MASLD, autoimmune hepatitis, and primary sclerosing cholangitis are particular concerns for disease recurrence in the transplanted liver, especially if underlying risk factors persist.

#391 · liver_disorders

What two scoring systems assess surgical risk before elective surgery in cirrhosis?

Answer

The Child-Turcotte-Pugh score and the MELD score. Patients with decompensated cirrhosis or a high MELD score should generally have elective surgery deferred or avoided.

#392 · liver_disorders

What distinguishes hepatopulmonary syndrome from portopulmonary hypertension?

Answer

Hepatopulmonary syndrome causes platypnea, worsening hypoxemia when upright that improves lying flat, from intrapulmonary vascular dilation. Portopulmonary hypertension involves pulmonary arterial hypertension and can complicate transplant timing if severe.

#393 · liver_disorders

How is hepatorenal syndrome diagnosed?

Answer

As a diagnosis of exclusion, a functional kidney injury from severe splanchnic vasodilation in advanced cirrhosis, made only after other causes of kidney injury have been ruled out.

#394 · liver_disorders

What is minimal hepatic encephalopathy, and why does it matter clinically?

Answer

A subtle cognitive impairment in cirrhosis that can affect driving safety and daily function despite a normal standard neurologic exam, making it easy to miss without specific testing.

#395 · liver_disorders

Why are patients with advanced cirrhosis more prone to bacterial and fungal infections?

Answer

Cirrhosis-associated immune dysfunction impairs normal immune surveillance, increasing susceptibility to both bacterial and fungal infections independent of any single specific organism.

#396 · pancreas_biliary_disorders

What are the two main causes of acute pancreatitis?

Answer

Gallstones and alcohol use together account for the large majority of cases. Other causes include hypertriglyceridemia, hypercalcemia, medications, and post-ERCP injury.

#397 · pancreas_biliary_disorders

Why is lipase preferred over amylase for diagnosing acute pancreatitis?

Answer

Lipase stays elevated longer after symptom onset and is more specific to the pancreas. Amylase is also produced by the salivary glands, making it less specific.

#398 · pancreas_biliary_disorders

What are the three diagnostic criteria for acute pancreatitis?

Answer

Typical epigastric pain radiating to the back, lipase or amylase at least three times normal, and characteristic imaging findings. At least two of the three must be present.

#399 · pancreas_biliary_disorders

Define mild acute pancreatitis under the Revised Atlanta Classification.

Answer

No organ failure and no local or systemic complications. This category has the best prognosis and typically resolves quickly.

#400 · pancreas_biliary_disorders

Define moderately severe acute pancreatitis under the Revised Atlanta Classification.

Answer

Organ failure lasting less than 48 hours, called transient organ failure, together with local or systemic complications.

#401 · pancreas_biliary_disorders

Define severe acute pancreatitis under the Revised Atlanta Classification.

Answer

Organ failure lasting 48 hours or more, called persistent organ failure, together with local or systemic complications. This carries the highest mortality.

#402 · pancreas_biliary_disorders

What is the single feature that separates severe from moderately severe acute pancreatitis?

Answer

Duration of organ failure. Persistent organ failure of 48 hours or more defines severe disease, while transient organ failure under 48 hours defines moderately severe disease.

#403 · pancreas_biliary_disorders

What is an acute peripancreatic fluid collection, and what can it become?

Answer

A fluid collection without a defined wall occurring in the first four weeks of acute pancreatitis. If it persists beyond four weeks and develops a wall, it becomes a pancreatic pseudocyst.

#404 · pancreas_biliary_disorders

How does walled-off necrosis differ from a pancreatic pseudocyst?

Answer

Walled-off necrosis contains solid necrotic debris within a defined wall, while a pseudocyst is purely fluid. Both are typically described only after four weeks.

#405 · pancreas_biliary_disorders

When should contrast-enhanced CT be obtained to assess pancreatic necrosis?

Answer

At 72 to 96 hours after symptom onset. Earlier imaging can underestimate the true extent of necrosis before it fully demarcates.

#406 · pancreas_biliary_disorders

What is the recommended fluid strategy in the first 24 hours of acute pancreatitis?

Answer

Aggressive intravenous isotonic crystalloid resuscitation to maintain pancreatic microcirculation and prevent hypovolemia-related complications.

#407 · pancreas_biliary_disorders

What feeding strategy is now preferred in acute pancreatitis over the old bowel rest approach?

Answer

Early oral feeding as tolerated, even in moderately severe disease. This is associated with fewer infectious complications than prolonged fasting.

#408 · pancreas_biliary_disorders

When should infected pancreatic necrosis be suspected?

Answer

When a patient with necrotizing pancreatitis fails to improve or worsens after one to two weeks, especially with new fever or rising white count.

#409 · pancreas_biliary_disorders

What is the step-up approach to infected pancreatic necrosis?

Answer

Starting with percutaneous or endoscopic drainage before escalating to more invasive surgical necrosectomy if the patient does not improve.

#410 · pancreas_biliary_disorders

When should cholecystectomy be performed in mild gallstone pancreatitis?

Answer

During the same hospital admission, since delaying it raises the risk of recurrent gallstone-related events before an elective later surgery date.

#411 · pancreas_biliary_disorders

What triglyceride level is generally considered a likely cause of acute pancreatitis?

Answer

Approximately 1000 mg/dL or greater. Lower levels are less likely to be the primary causative factor on their own.

#412 · pancreas_biliary_disorders

What is the classic mechanism by which alcohol causes acute pancreatitis?

Answer

Direct toxic injury to acinar cells combined with premature intracellular activation of digestive enzymes, distinct from the mechanical obstruction seen in gallstone disease.

#413 · pancreas_biliary_disorders

What is pancreas divisum?

Answer

A congenital variant where the dorsal and ventral pancreatic ducts fail to fuse, so most drainage occurs through the smaller minor papilla instead of the major papilla.

#414 · pancreas_biliary_disorders

What is annular pancreas?

Answer

A congenital anomaly where a ring of pancreatic tissue encircles the duodenum, which can cause duodenal obstruction in infancy or later in life.

#415 · pancreas_biliary_disorders

Why are choledochal cysts treated with surgical excision rather than simple drainage?

Answer

Because they carry a meaningfully increased long-term risk of cholangiocarcinoma, so removing the cyst addresses this cancer risk directly.

#416 · pancreas_biliary_disorders

What structure forms when the pancreatic duct and common bile duct join?

Answer

The ampulla of Vater, which then drains into the duodenum through the major papilla, controlled by the sphincter of Oddi.

#417 · pancreas_biliary_disorders

What is biliary colic?

Answer

Steady right upper quadrant or epigastric pain from transient cystic duct obstruction by a stone, often after fatty meals, resolving over a few hours without fever.

#418 · pancreas_biliary_disorders

What are the classic ultrasound findings of acute cholecystitis?

Answer

Gallbladder wall thickening greater than 3 millimeters, pericholecystic fluid, gallbladder distension, and a sonographic Murphy sign.

#419 · pancreas_biliary_disorders

What is the recommended timing for cholecystectomy in acute cholecystitis?

Answer

Early laparoscopic cholecystectomy, ideally within 72 hours of symptom onset, along with antibiotics covering enteric organisms.

#420 · pancreas_biliary_disorders

What is acalculous cholecystitis, and who gets it?

Answer

Cholecystitis without stones, seen in critically ill patients from gallbladder ischemia and bile stasis. It carries a higher risk of gangrene and perforation than calculous disease.

#421 · pancreas_biliary_disorders

What is the temporizing treatment for acute cholecystitis in an unstable patient?

Answer

Percutaneous cholecystostomy tube placement, which drains the gallbladder without the physiologic stress of surgery.

#422 · pancreas_biliary_disorders

What does a HIDA scan showing nonvisualization of the gallbladder imply?

Answer

Persistent cystic duct obstruction, supporting the diagnosis of acute cholecystitis when ultrasound findings are equivocal.

#423 · pancreas_biliary_disorders

What features raise concern for malignancy in a gallbladder polyp?

Answer

Polyp size larger than 1 centimeter, documented interval growth, or occurrence in a patient with primary sclerosing cholangitis.

#424 · pancreas_biliary_disorders

What is Mirizzi syndrome?

Answer

Extrinsic compression of the common hepatic duct by a stone impacted in the cystic duct or gallbladder neck, causing obstructive jaundice.

#425 · pancreas_biliary_disorders

What is gallstone ileus?

Answer

Mechanical small bowel obstruction from a large gallstone that erodes through a cholecystoenteric fistula and lodges, most often at the terminal ileum.

#426 · pancreas_biliary_disorders

What lab pattern is expected with choledocholithiasis?

Answer

A cholestatic pattern, with alkaline phosphatase and bilirubin rising more than the transaminases.

#427 · pancreas_biliary_disorders

When is ERCP preferred over MRCP for a suspected bile duct stone?

Answer

When the pretest probability of a stone is already high, since ERCP allows both diagnosis and immediate therapeutic sphincterotomy and stone extraction.

#428 · pancreas_biliary_disorders

What is Charcot triad, and what does Reynolds pentad add to it?

Answer

Charcot triad is fever, jaundice, and right upper quadrant pain from cholangitis. Reynolds pentad adds altered mental status and hypotension, marking more severe disease.

#429 · pancreas_biliary_disorders

What is the emergency management of acute cholangitis?

Answer

Intravenous antibiotics with prompt biliary drainage, most often by ERCP, since infected bile under pressure needs urgent decompression.

#430 · pancreas_biliary_disorders

What disease is primary sclerosing cholangitis strongly associated with?

Answer

Inflammatory bowel disease, especially ulcerative colitis. It also raises the lifetime risk of cholangiocarcinoma and colorectal cancer.

#431 · pancreas_biliary_disorders

What is the classic MRCP finding in primary sclerosing cholangitis?

Answer

A beaded appearance of alternating strictures and dilations throughout the intrahepatic and extrahepatic bile ducts.

#432 · pancreas_biliary_disorders

Does ursodeoxycholic acid change the natural history of primary sclerosing cholangitis?

Answer

No. It can improve liver biochemistries but has not been shown to change disease progression at high doses, so it is not routinely recommended for that purpose.

#433 · pancreas_biliary_disorders

What antibody is characteristic of primary biliary cholangitis?

Answer

Antimitochondrial antibody, positive in the great majority of affected patients, most often middle-aged women.

#434 · pancreas_biliary_disorders

Is ursodeoxycholic acid effective first-line therapy in primary biliary cholangitis?

Answer

Yes. Unlike in primary sclerosing cholangitis, it is first-line therapy here and has been shown to improve long-term outcomes.

#435 · pancreas_biliary_disorders

What causes post-ERCP pancreatitis?

Answer

Mechanical or hydrostatic injury to the pancreatic duct during cannulation and sphincterotomy, presenting with typical pancreatitis pain and lipase elevation shortly after the procedure.

#436 · pancreas_biliary_disorders

What is the most common identifiable cause of chronic pancreatitis in adults?

Answer

Chronic alcohol use. A meaningful share of cases remain idiopathic, and genetic causes should be considered in younger patients.

#437 · pancreas_biliary_disorders

Which genetic mutations should be considered in a young patient with chronic pancreatitis and no typical risk factors?

Answer

PRSS1, SPINK1, and CFTR mutations.

#438 · pancreas_biliary_disorders

What is the most sensitive test for early chronic pancreatitis, and what limits its use?

Answer

Secretin stimulation testing is the most sensitive, but it requires specialized equipment and is not widely available outside specialized centers.

#439 · pancreas_biliary_disorders

What is a key limitation of fecal elastase testing?

Answer

It only becomes abnormal after a substantial portion of exocrine function is already lost, so it can miss early or mild disease.

#440 · pancreas_biliary_disorders

What type of diabetes results from chronic pancreatitis, and why?

Answer

Type 3c or pancreatogenic diabetes, from progressive loss of islet cell mass as functional pancreatic tissue is destroyed over time.

#441 · pancreas_biliary_disorders

Has pancreatic enzyme replacement therapy been shown to reduce pain in chronic pancreatitis?

Answer

Not reliably in most trials, even though enzyme therapy remains important for treating malabsorption.

#442 · pancreas_biliary_disorders

When is surgery like lateral pancreaticojejunostomy used in chronic pancreatitis?

Answer

In patients who fail endoscopic therapy or who have diffuse ductal disease not suited to a focal endoscopic approach.

#443 · pancreas_biliary_disorders

What malignancy risk is significantly elevated in chronic pancreatitis, especially hereditary forms?

Answer

Pancreatic adenocarcinoma. Any new or changed pain pattern in a chronic pancreatitis patient warrants imaging to exclude a new mass.

#444 · pancreas_biliary_disorders

At what functional threshold does exocrine pancreatic insufficiency typically become clinically apparent?

Answer

Once functional capacity falls below approximately 10 percent of normal, reflecting the pancreas's substantial reserve capacity.

#445 · pancreas_biliary_disorders

Why should pancreatic enzyme replacement therapy be taken with meals rather than before or after eating?

Answer

Enzymes need to be present in the duodenum at the same time as food to properly digest it, so timing with meals maximizes effectiveness.

#446 · nutrition

Where is vitamin B12 absorbed?

Answer

In the terminal ileum, after binding to intrinsic factor made by gastric parietal cells. Terminal ileal disease or resection, or loss of intrinsic factor from autoimmune gastritis, causes B12 deficiency with macrocytic anemia and neurologic symptoms.

#447 · nutrition

Why does methylmalonic acid distinguish B12 from folate deficiency?

Answer

B12 is a required cofactor for the enzyme that clears methylmalonic acid, so it rises only in B12 deficiency. Folate deficiency does not raise methylmalonic acid, while homocysteine rises in both.

#448 · nutrition

Why can long term PPI use cause iron deficiency?

Answer

Iron absorption through the duodenal DMT1 transporter requires an acidic environment. Acid suppression from a proton pump inhibitor impairs this process and can contribute to iron deficiency over time.

#449 · nutrition

What is acrodermatitis and what deficiency causes it?

Answer

A scaly rash around the mouth, hands, and feet caused by zinc deficiency. It is seen after bariatric surgery and in Crohn disease, along with poor wound healing and hair loss.

#450 · nutrition

Why does excess zinc supplementation cause copper deficiency?

Answer

Zinc competes with copper for intestinal absorption. Excess zinc intake, common after bariatric surgery, can cause copper deficiency presenting with sideroblastic anemia and a B12-deficiency-like neurologic syndrome.

#451 · nutrition

Which four vitamins are lost together in fat malabsorption?

Answer

Vitamins A, D, E, and K, since all four require bile acids and pancreatic enzymes for absorption. Any cause of fat malabsorption can deplete all four simultaneously.

#452 · nutrition

What does vitamin A excess cause?

Answer

Pseudotumor cerebri (headache, blurred vision, papilledema without a mass) and skin changes, usually from over supplementation. Vitamin A deficiency instead causes night blindness.

#453 · nutrition

What is ARFID and how does it differ from anorexia nervosa?

Answer

Avoidant/restrictive food intake disorder is a feeding disorder driven by sensory sensitivity, fear of choking, or low interest in food, not by body image concerns. Anorexia nervosa specifically requires a body image disturbance that ARFID lacks.

#454 · nutrition

What are the three broad mechanisms of malabsorption?

Answer

Maldigestion (insufficient enzymes or bile, as in chronic pancreatitis), mucosal disease (damaged absorptive surface, as in celiac disease), and anatomic or motility problems (reduced contact time, as in short bowel syndrome or scleroderma).

#455 · nutrition

What does a low fecal elastase indicate?

Answer

Pancreatic exocrine insufficiency. It is a simple spot stool test that has largely replaced the 72 hour fecal fat collection as a screening tool for pancreatic causes of steatorrhea.

#456 · nutrition

What does the D-xylose test evaluate and why is it useful?

Answer

It tests small bowel mucosal absorptive capacity without requiring pancreatic enzymes. A low result with a normal fecal elastase points to a mucosal or bacterial cause of malabsorption rather than pancreatic insufficiency.

#457 · nutrition

Why can celiac disease present without diarrhea?

Answer

It can present subtly with isolated iron deficiency anemia or osteoporosis from malabsorption of iron and calcium/vitamin D, even without GI symptoms, warranting a low threshold for tissue transglutaminase IgA testing.

#458 · nutrition

What GLP-1 side effect matters most before endoscopy?

Answer

Delayed gastric emptying, which raises aspiration risk during sedation. Many centers require an extended liquid diet the day before endoscopy in patients on GLP-1 or GIP/GLP-1 receptor agonists.

#459 · nutrition

What is the difference between an intragastric balloon and endoscopic sleeve gastroplasty?

Answer

A balloon is a temporary space-occupying device removed after a set period. Sleeve gastroplasty uses endoscopic sutures to permanently reduce gastric volume without surgical resection.

#460 · nutrition

Which nutrients require lifelong monitoring after Roux-en-Y gastric bypass?

Answer

Iron, vitamin B12, folate, calcium, and the fat soluble vitamins (A, D, E, K), since bypassed anatomy predisposes to deficiency in all of these long term.

#461 · nutrition

What is the GLIM approach to diagnosing malnutrition?

Answer

It combines phenotypic criteria (weight loss, low BMI, reduced muscle mass) with etiologic criteria (reduced intake or an inflammatory disease state), so a normal-weight patient can still be diagnosed as malnourished.

#462 · nutrition

Why does sarcopenia matter independently of Child-Pugh or MELD score?

Answer

Sarcopenia (loss of skeletal muscle mass and strength) independently predicts worse outcomes after transplant or surgery in cirrhosis, regardless of the standard liver severity scores, so it should be assessed separately.

#463 · nutrition

What is refeeding syndrome and what electrolyte is most critical?

Answer

A dangerous shift of phosphate, potassium, and magnesium into cells triggered by an insulin surge when nutrition is reintroduced too quickly after starvation. Severe hypophosphatemia is the most critical finding, risking arrhythmia and respiratory failure.

#464 · nutrition

How is refeeding syndrome prevented?

Answer

Start nutrition at a low caloric rate, replete electrolytes before and during feeding, and monitor phosphate, potassium, and magnesium daily during the first week of refeeding.

#465 · nutrition

When is jejunal (post-pyloric) feeding preferred over gastric feeding?

Answer

In patients with severe gastroparesis, high aspiration risk, or acute pancreatitis, since bypassing the stomach reduces the volume of retained contents available for aspiration.

#466 · other_disorders

What are the two subtypes of microscopic colitis and how do they differ histologically?

Answer

Collagenous colitis shows a thickened subepithelial collagen band. Lymphocytic colitis shows increased intraepithelial lymphocytes without the thickened band. Both cause chronic watery diarrhea with a grossly normal colon.

#467 · other_disorders

Which medications are most strongly associated with microscopic colitis?

Answer

NSAIDs, proton pump inhibitors, and SSRIs. Discontinuing the offending agent is an important adjunct to budesonide therapy.

#468 · other_disorders

Why is budesonide preferred over prednisone for microscopic colitis?

Answer

Budesonide has high first pass hepatic metabolism, which limits systemic corticosteroid side effects compared with prednisone, while still effectively treating the colonic inflammation.

#469 · other_disorders

Why might colonic biopsies miss microscopic colitis?

Answer

Histologic changes can be patchy and sometimes absent in the rectosigmoid colon, so random biopsies should be taken from multiple colonic segments, not the rectum alone.

#470 · other_disorders

How do you calculate the stool osmotic gap?

Answer

290 minus two times the sum of stool sodium and potassium. A high gap suggests an osmotic cause of diarrhea, while a low gap suggests a secretory cause.

#471 · other_disorders

What defines secretory versus osmotic diarrhea?

Answer

Secretory diarrhea persists with fasting and has a normal or low osmotic gap (e.g. VIPoma, toxins). Osmotic diarrhea improves with fasting and has an elevated osmotic gap (e.g. lactose, magnesium laxatives).

#472 · other_disorders

Why does cholecystectomy sometimes cause chronic diarrhea?

Answer

Without the gallbladder buffering bile release, continuous bile flow into the duodenum can overwhelm ileal bile acid reabsorption capacity, producing bile acid diarrhea.

#473 · other_disorders

What does fecal calprotectin indicate and what is its main use?

Answer

It rises with neutrophil migration into the gut lumen during inflammation. It helps distinguish inflammatory bowel disease from a functional disorder like irritable bowel syndrome before pursuing colonoscopy.

#474 · other_disorders

What predisposes patients to small intestinal bacterial overgrowth (SIBO)?

Answer

Anatomic stagnant loops (prior surgery, strictures, blind loops) and conditions that slow motility (scleroderma, diabetic neuropathy, opioid use), which allow bacteria to accumulate in the small bowel.

#475 · other_disorders

How does SIBO cause fat malabsorption?

Answer

Bacteria deconjugate bile acids, impairing micelle formation needed for fat absorption, leading to steatorrhea and fat soluble vitamin deficiency.

#476 · other_disorders

How is SIBO diagnosed and treated?

Answer

Diagnosed with a glucose or lactulose hydrogen (or hydrogen-methane) breath test showing an early hydrogen rise. Treated with a poorly absorbed antibiotic like rifaximin, plus correction of the underlying cause to prevent relapse.

#477 · other_disorders

Why do internal hemorrhoids cause painless bleeding while external hemorrhoids are painful?

Answer

Internal hemorrhoids lie above the dentate line, an area without somatic pain innervation. External hemorrhoids lie below the dentate line, innervated by somatic nerves, so thrombosis there is painful.

#478 · other_disorders

What is first line treatment for a chronic anal fissure?

Answer

A topical agent that relaxes the internal anal sphincter, such as nifedipine or nitroglycerin, addressing the reduced local blood flow thought to cause chronicity. Sphincterotomy is reserved for medical failures.

#479 · other_disorders

When should recurrent perianal abscesses prompt evaluation for Crohn disease?

Answer

When abscesses are recurrent or complex, especially with associated abdominal pain or weight loss, warranting pelvic MRI and colonoscopy to evaluate for underlying Crohn disease.

#480 · other_disorders

What is the most common cause of small bowel obstruction in adults?

Answer

Postoperative adhesions from prior abdominal surgery, followed by hernias and malignancy.

#481 · other_disorders

What is the coffee bean sign and what does it indicate?

Answer

A dilated loop of colon bending back on itself on abdominal x-ray, classic for sigmoid volvulus, typically seen in elderly or institutionalized patients with chronic constipation.

#482 · other_disorders

How does adult intussusception differ from pediatric intussusception?

Answer

Adult intussusception almost always has an identifiable lead point (polyp, tumor, Meckel diverticulum) and generally requires surgical resection, unlike pediatric cases which are often idiopathic and can be reduced.

#483 · other_disorders

Why is a closed loop obstruction especially dangerous?

Answer

A bowel segment obstructed at both proximal and distal points cannot decompress in either direction, creating a high risk of rapid ischemia and perforation, requiring urgent surgical evaluation.

#484 · other_disorders

What is an obturator hernia and who typically gets it?

Answer

A hernia through the obturator canal, classically in thin, elderly women, causing medial thigh pain from obturator nerve compression, often without a palpable external bulge.

#485 · other_disorders

Why are femoral hernias higher risk than inguinal hernias?

Answer

Femoral hernias pass through a narrow, rigid canal below the inguinal ligament, making them more prone to incarceration and strangulation. They are also more common in women.

#486 · other_disorders

What is the classic presentation of acute mesenteric ischemia and its typical cause?

Answer

Sudden severe abdominal pain out of proportion to exam findings, classically from embolic occlusion of the superior mesenteric artery in a patient with atrial fibrillation.

#487 · other_disorders

What is chronic mesenteric ischemia (intestinal angina)?

Answer

Postprandial abdominal pain from atherosclerotic narrowing of at least two of the three mesenteric vessels, leading to food fear and weight loss over time.

#488 · other_disorders

How does ischemic colitis differ from acute mesenteric ischemia?

Answer

Ischemic colitis is usually nonocclusive, related to a low flow state, and affects watershed areas like the splenic flexure. It is generally managed supportively, unlike the surgical emergency of acute mesenteric ischemia.

#489 · other_disorders

What is Heyde syndrome?

Answer

The association between aortic stenosis and bleeding colonic angioectasias, caused by acquired von Willebrand factor deficiency from shear stress across the stenotic valve. Bleeding often improves after valve replacement.

#490 · other_disorders

What is the rule of twos for Meckel diverticulum?

Answer

Occurs in about 2 percent of the population, usually within 2 feet of the ileocecal valve, and often contains ectopic gastric or pancreatic tissue that can cause painless bleeding.

#491 · other_disorders

How has management of uncomplicated appendicitis changed?

Answer

A trial of antibiotics alone is now an accepted alternative to appendectomy in appropriately selected patients without perforation or abscess, though surgery remains standard for complicated disease.

#492 · other_disorders

How is a periappendiceal abscess typically managed?

Answer

Percutaneous drainage plus antibiotics first, followed by interval appendectomy weeks later, since immediate surgery in a densely inflamed field carries higher complication rates.

#493 · other_disorders

Why do elderly patients with appendicitis often present atypically?

Answer

They often have blunted inflammatory responses (less fever, less leukocytosis), which delays diagnosis and increases the risk of presenting with perforation compared with younger patients.

#494 · other_disorders

Why does loss of the ileocecal valve worsen short bowel syndrome?

Answer

The valve normally slows small bowel transit and limits reflux of colonic bacteria into the small intestine. Its loss accelerates transit and promotes bacterial overgrowth, worsening malabsorption.

#495 · other_disorders

What is the difference between early and late dumping syndrome?

Answer

Early dumping occurs within 30 minutes from rapid fluid shift into the bowel after hyperosmolar food enters quickly, causing tachycardia and cramping. Late dumping occurs 1 to 3 hours later from reactive hypoglycemia after an exaggerated insulin surge.

#496 · other_disorders

Why can internal hernias after Roux-en-Y bypass be missed on imaging?

Answer

They can reduce spontaneously, so a CT scan obtained during an asymptomatic period can be falsely reassuring, requiring a high index of suspicion despite normal imaging.

#497 · other_disorders

What is the difference between diverticulosis and diverticulitis?

Answer

Diverticulosis is asymptomatic outpouching of the colonic wall. Diverticulitis is inflammation or micro-perforation of a diverticulum causing pain, fever, and leukocytosis.

#498 · other_disorders

When is colonoscopy performed after acute diverticulitis, and why the timing?

Answer

Six to eight weeks after the episode resolves, to exclude an underlying malignancy. It is never performed during the acute phase because of perforation risk.

#499 · other_disorders

What is a colovesical fistula and its classic symptom triad?

Answer

A fistula between the colon and bladder, most often from diverticulitis, causing recurrent UTIs, pneumaturia, and fecaluria. It typically requires elective surgical resection.

#500 · other_disorders

What is the Carnett sign and how is it interpreted?

Answer

A bedside test for abdominal wall pain: the examiner palpates the point of maximal tenderness while the patient tenses the abdominal muscles. Pain that worsens or stays the same suggests an abdominal wall source; pain that improves suggests an intra-abdominal source.